

ERD vật lý / master / organization

WMS / BA 2.0 / 27-09-2026 / DRAFT

organization
PK id : uuid
code : varchar(40)
name : varchar(200)
timezone : varchar(80)
currency : char(3)

ERD vật lý / iam / app_user

WMS / BA 2.0 / 27-09-2026 / DRAFT

app_user
PK id : uuid
username : varchar(100)
display_name : varchar(200)
password_hash : text
is_active : boolean
auth_version : integer
created_at : timestamptz

ERD vật lý / iam / role

WMS / BA 2.0 / 27-09-2026 / DRAFT

role
PK id : uuid
code : varchar(60)
name : varchar(160)
is_active : boolean

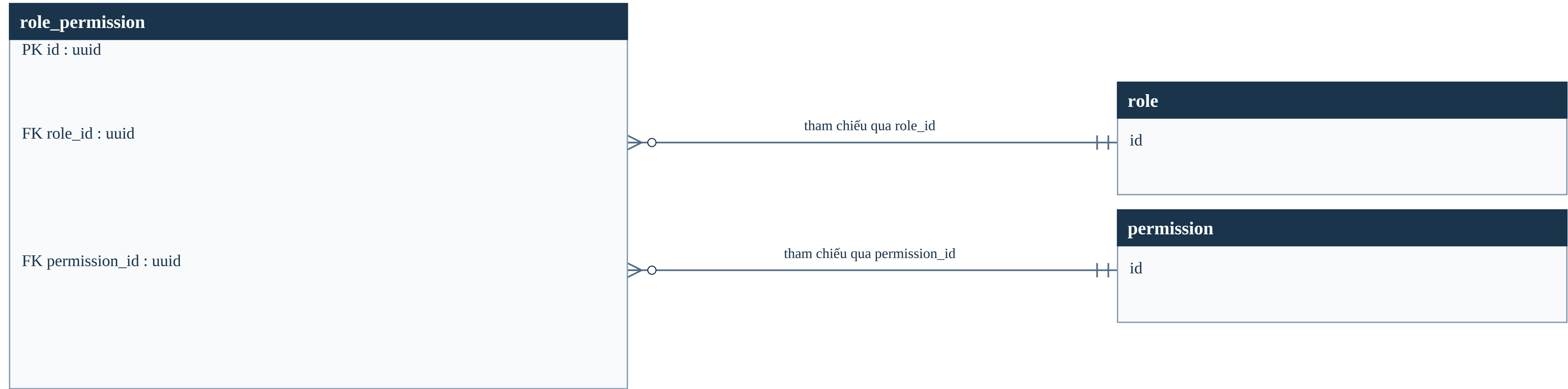
ERD vật lý / iam / permission

WMS / BA 2.0 / 27-09-2026 / DRAFT

permission
PK id : uuid
code : varchar(100)
scope_kind : varchar(20)
description : text

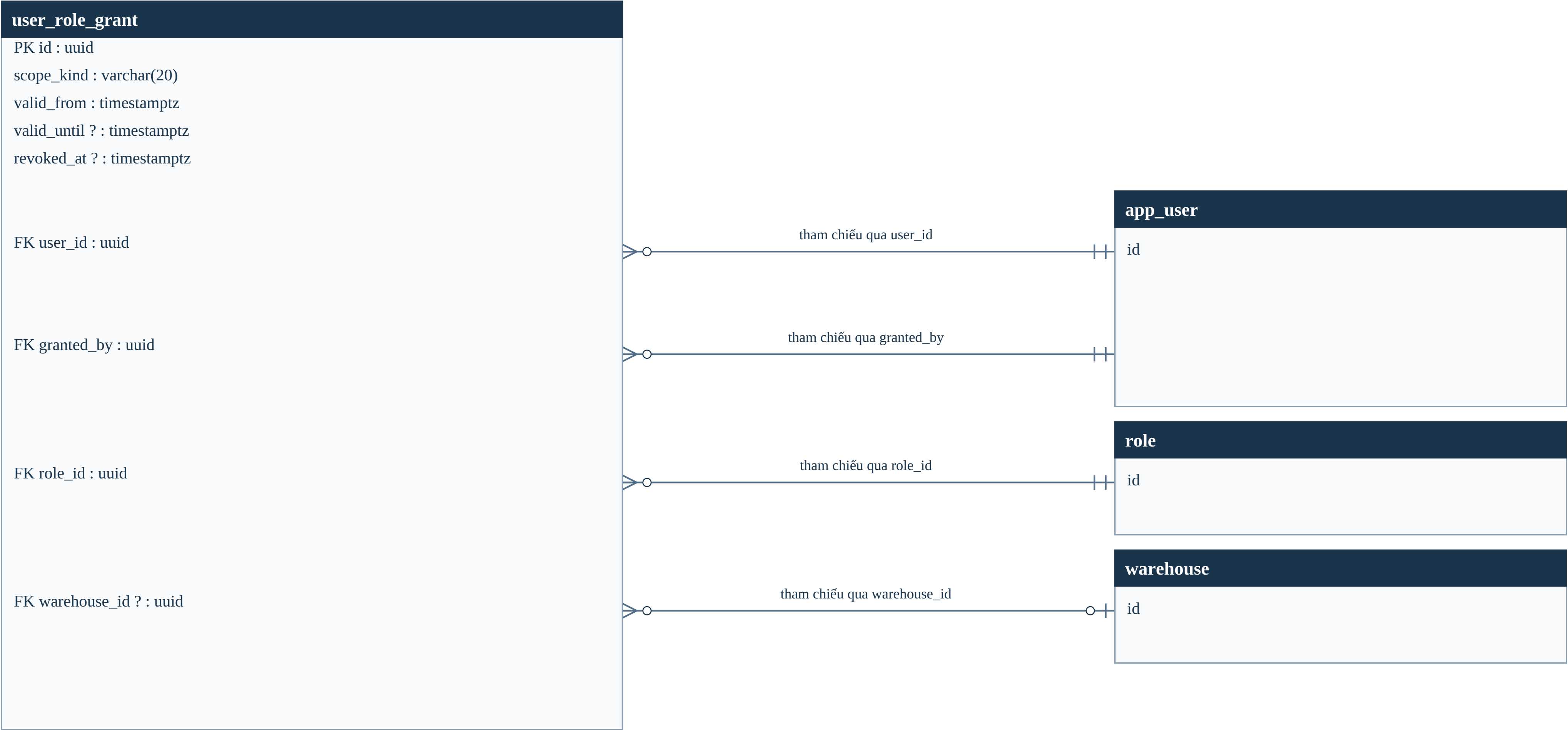
ERD vật lý / iam / role_permission

WMS / BA 2.0 / 27-09-2026 / DRAFT



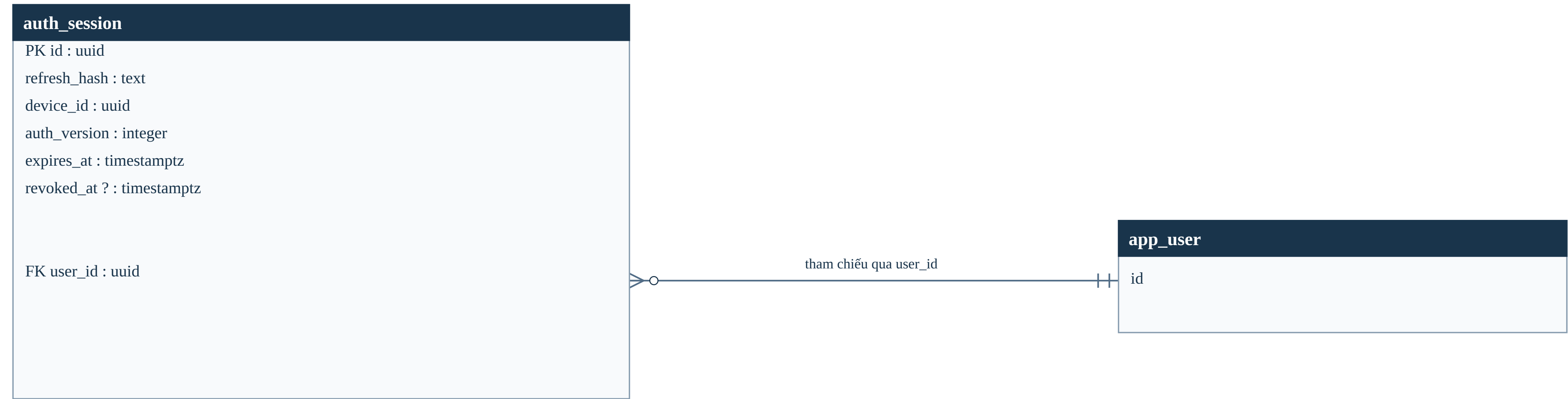
ERD vật lý / iam / user_role_grant

WMS / BA 2.0 / 27-09-2026 / DRAFT



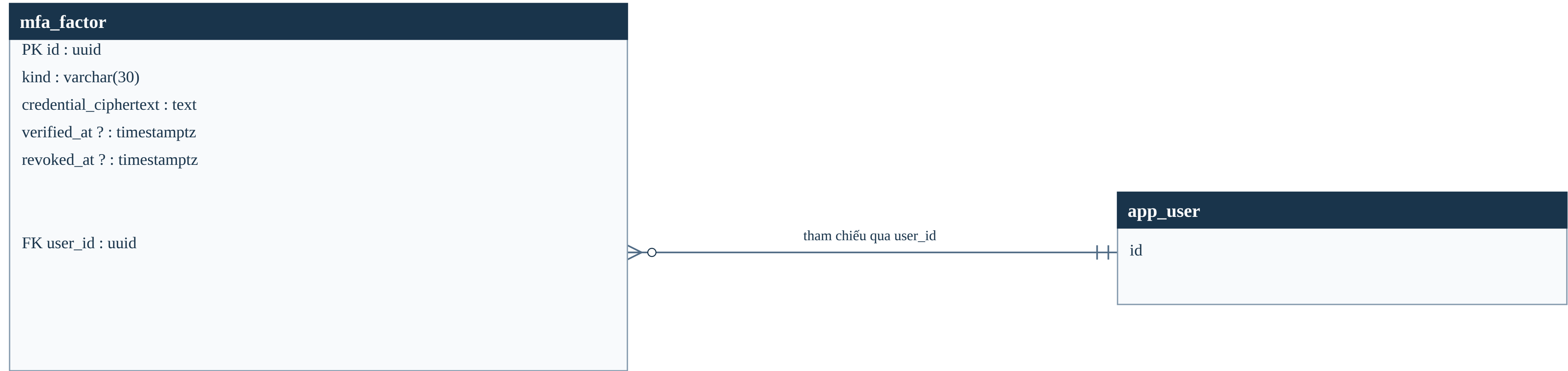
ERD vật lý / iam / auth_session

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ERD vật lý / iam / mfa_factor

WMS / BA 2.0 / 27-09-2026 / DRAFT



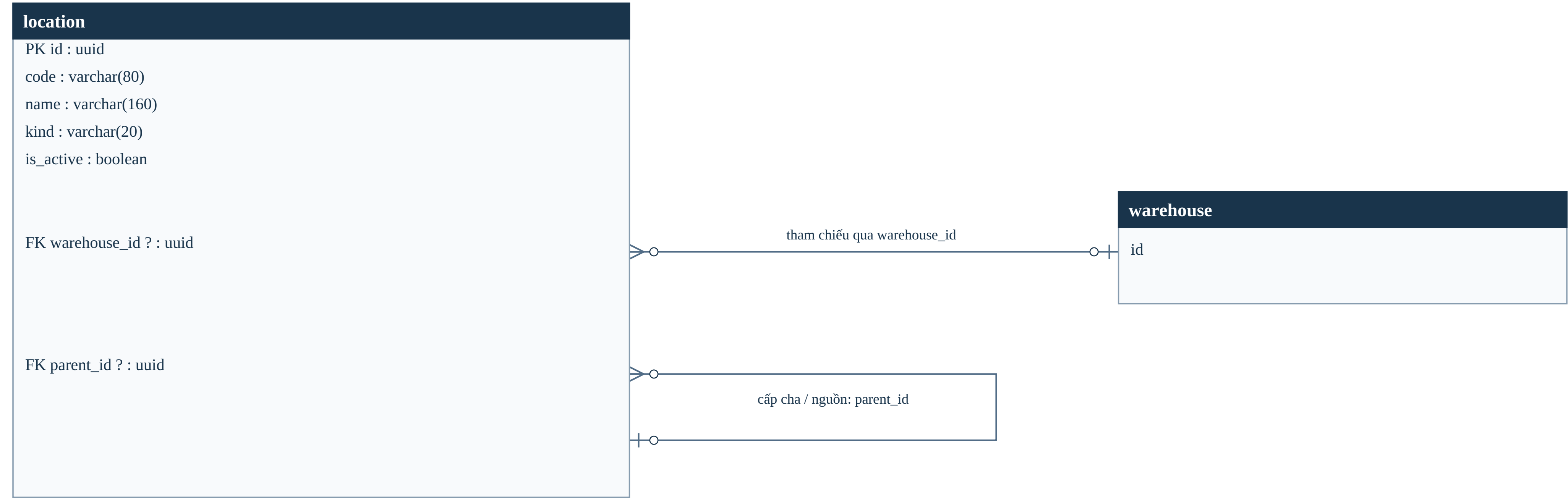
ERD vật lý / master / warehouse

WMS / BA 2.0 / 27-09-2026 / DRAFT

warehouse
PK id : uuid
code : varchar(40)
name : varchar(160)
address ? : text
is_active : boolean

ERD vật lý / master / location

WMS / BA 2.0 / 27-09-2026 / DRAFT



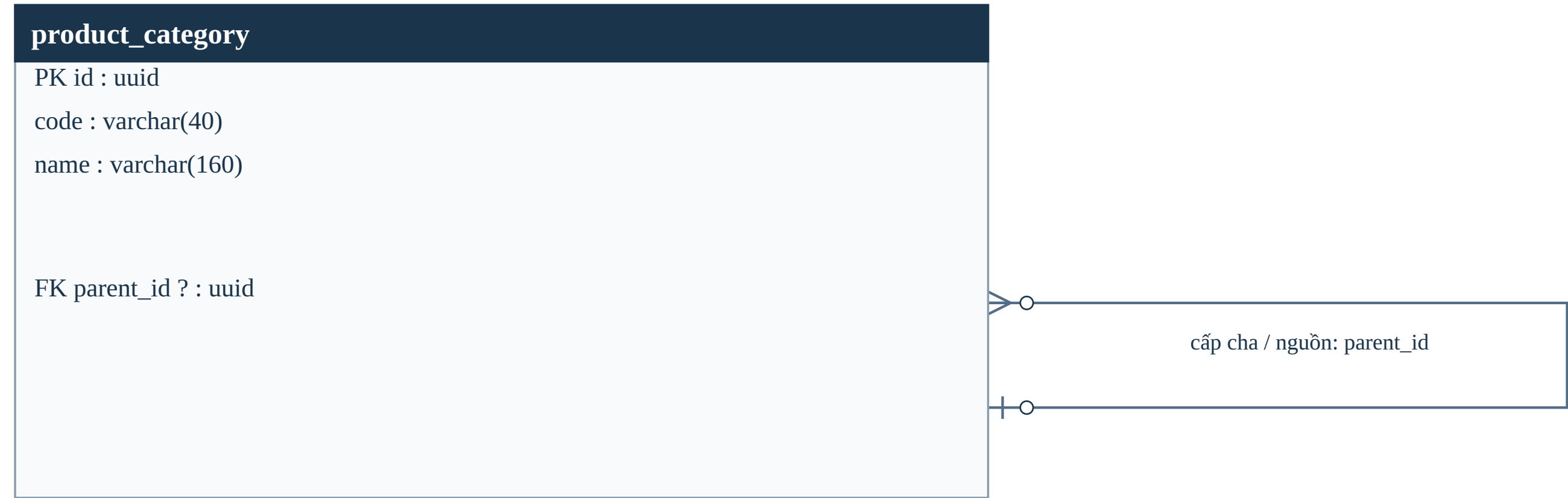
ERD vật lý / master / uom

WMS / BA 2.0 / 27-09-2026 / DRAFT

uom
PK id : uuid
code : varchar(30)
name : varchar(80)
decimal_places : integer

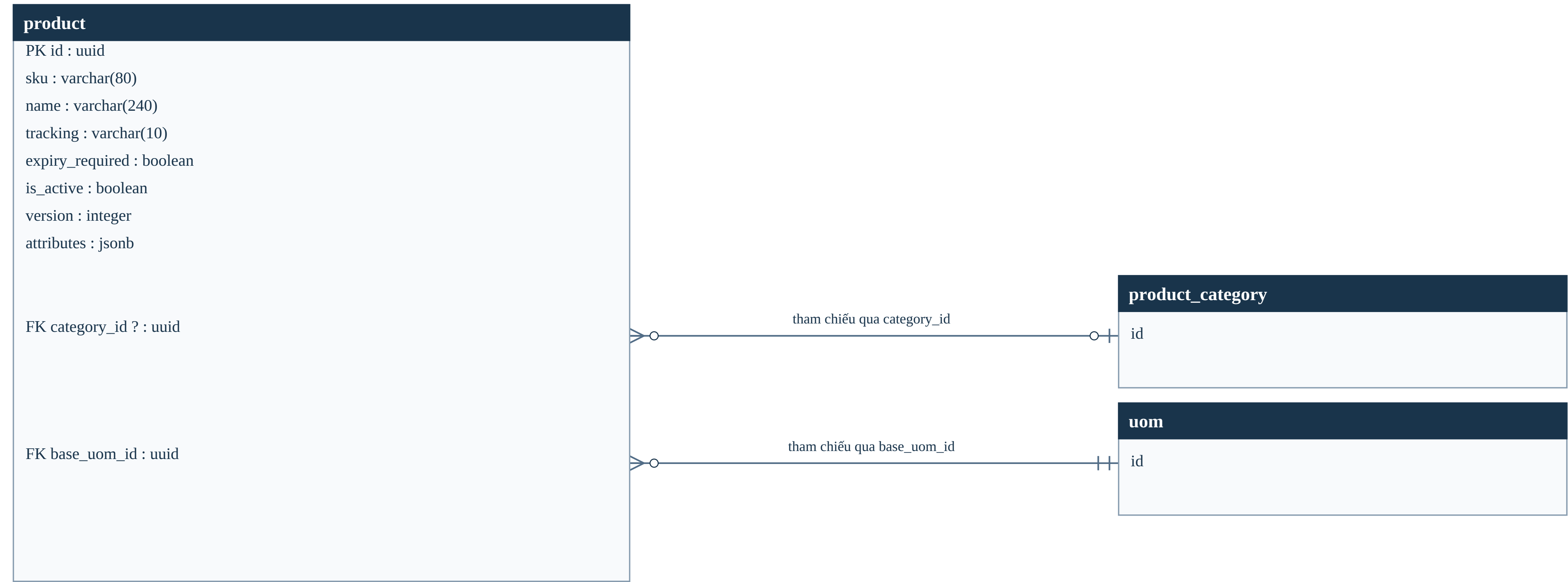
ERD vật lý / master / product_category

WMS / BA 2.0 / 27-09-2026 / DRAFT



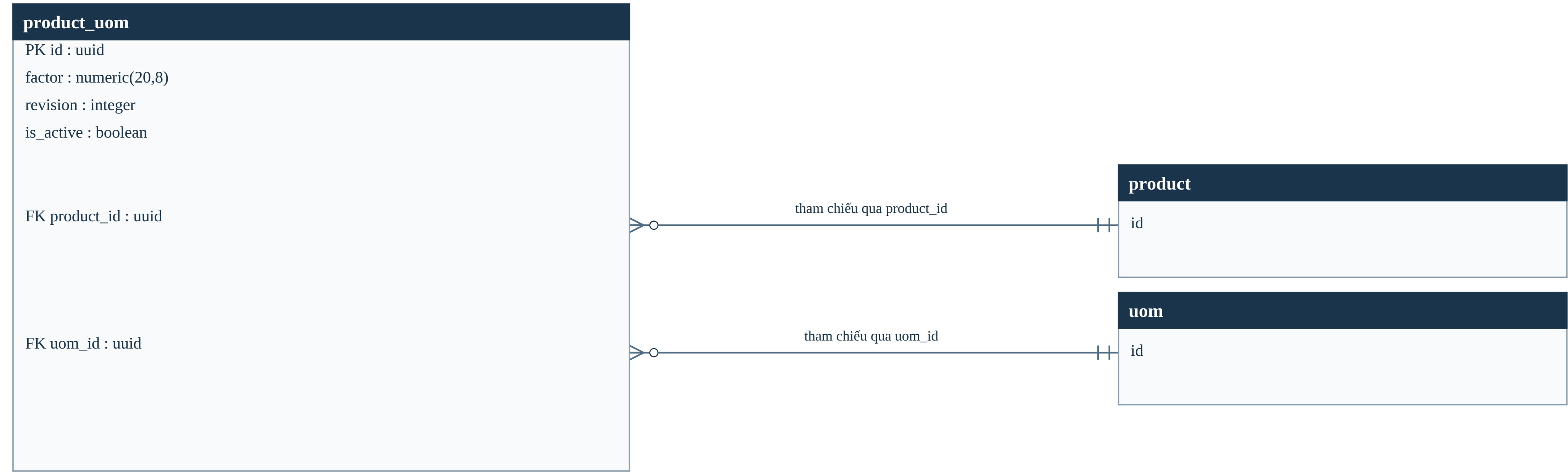
ERD vật lý / master / product

WMS / BA 2.0 / 27-09-2026 / DRAFT



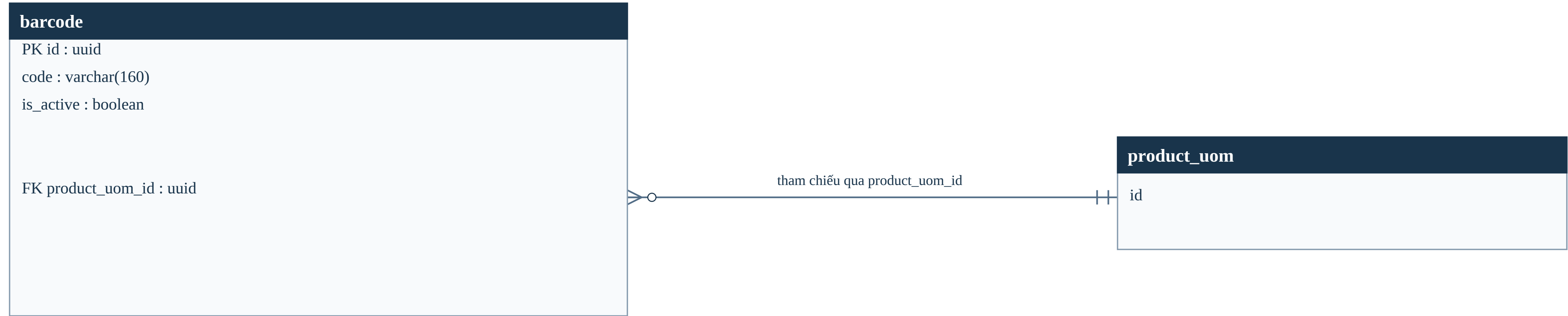
ERD vật lý / master / product_uom

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ERD vật lý / master / barcode

WMS / BA 2.0 / 27-09-2026 / DRAFT



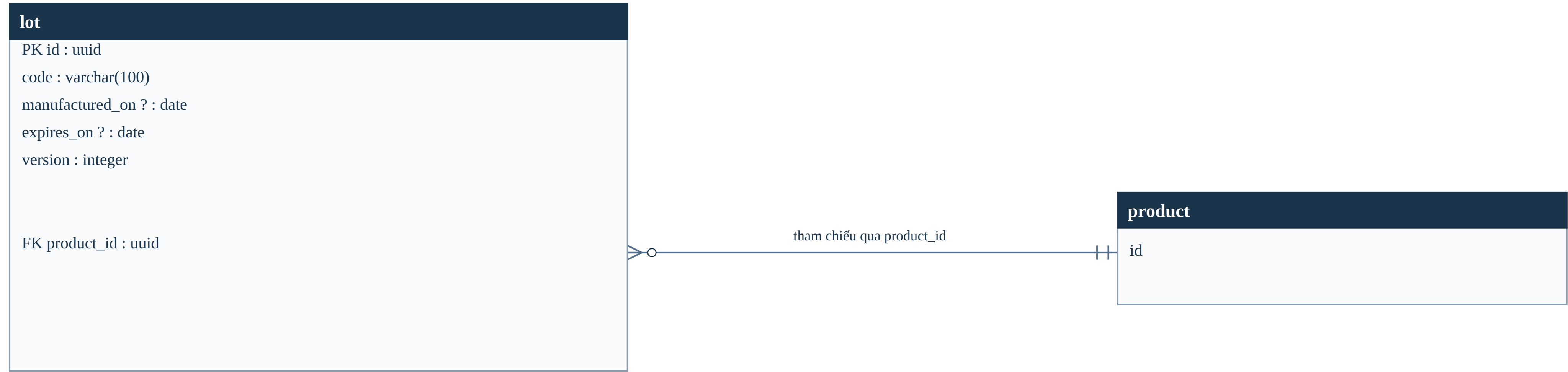
ERD vật lý / master / partner

WMS / BA 2.0 / 27-09-2026 / DRAFT

partner
PK id : uuid
code : varchar(60)
name : varchar(240)
is_customer : boolean
is_supplier : boolean
tax_code ? : varchar(40)
address ? : text
is_active : boolean

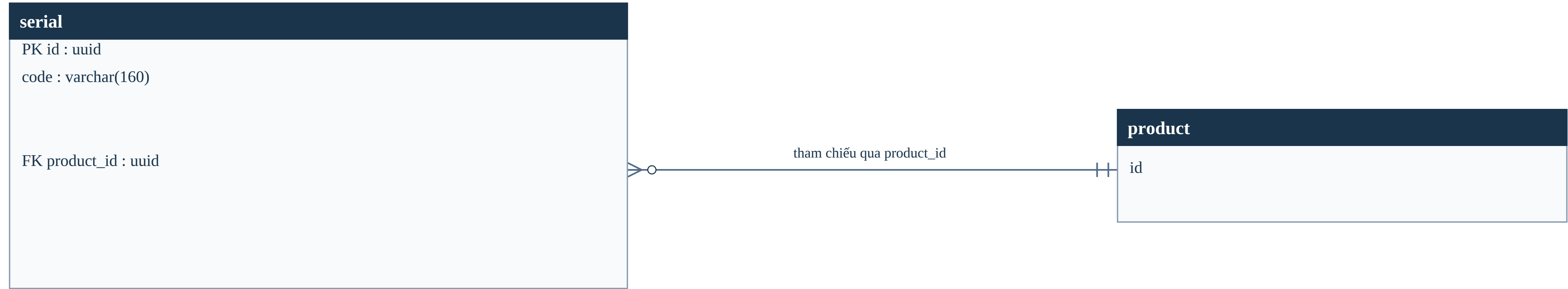
ERD vật lý / inventory / lot

WMS / BA 2.0 / 27-09-2026 / DRAFT



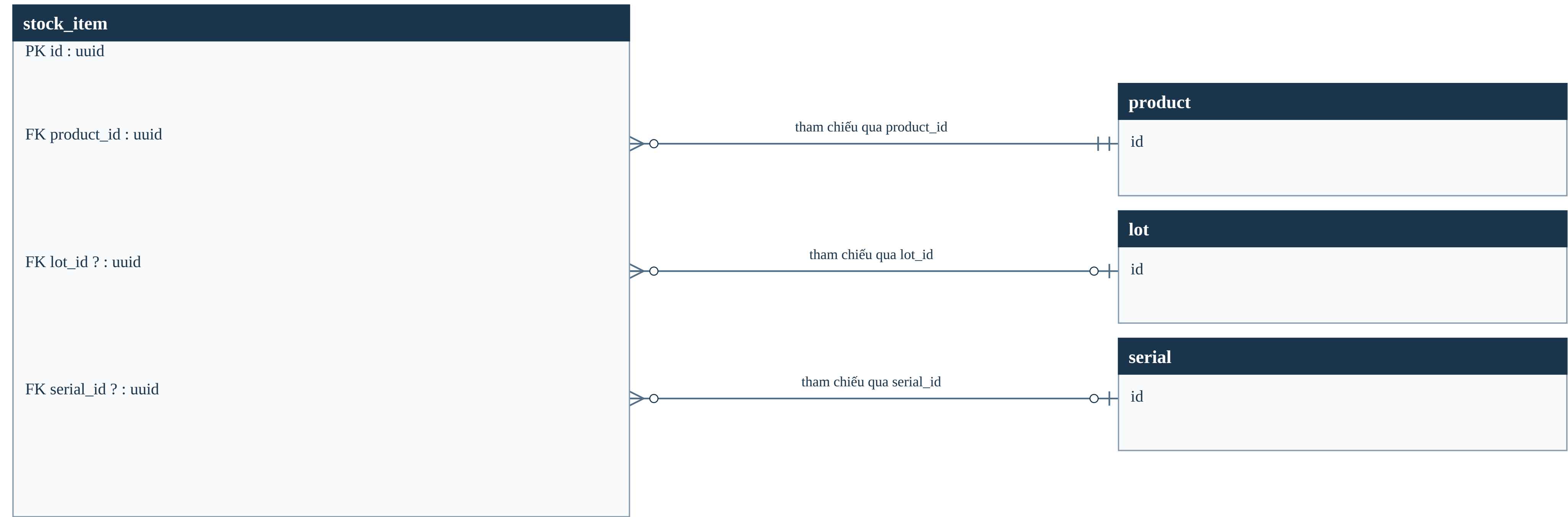
ERD vật lý / inventory / serial

WMS / BA 2.0 / 27-09-2026 / DRAFT



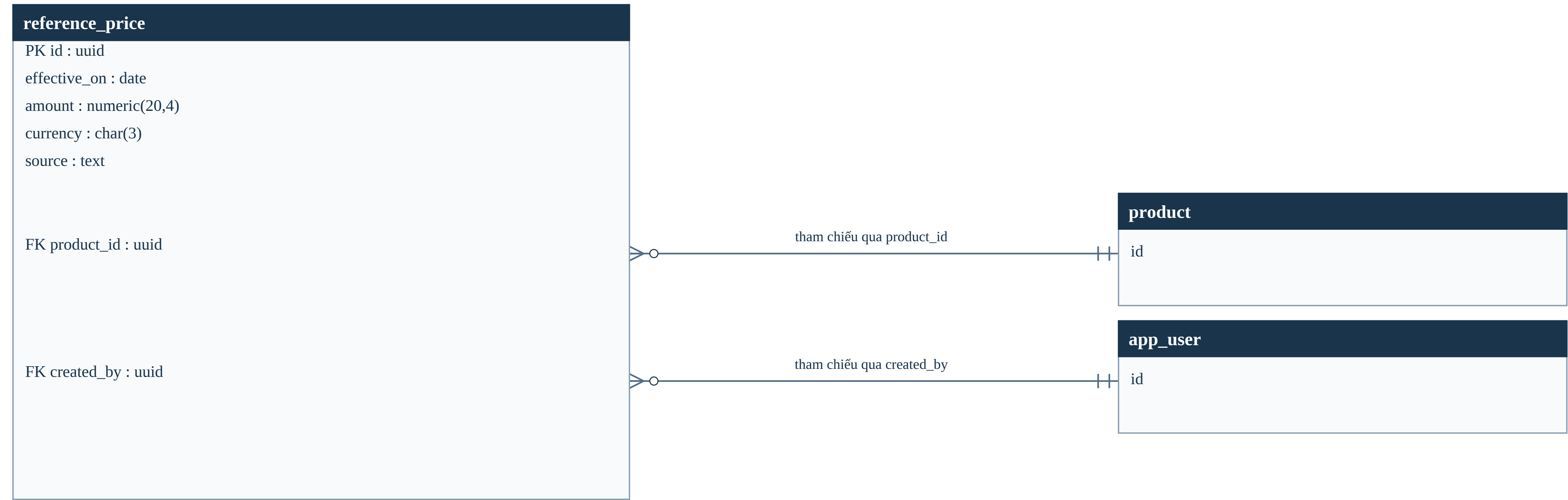
ERD vật lý / inventory / stock_item

WMS / BA 2.0 / 27-09-2026 / DRAFT



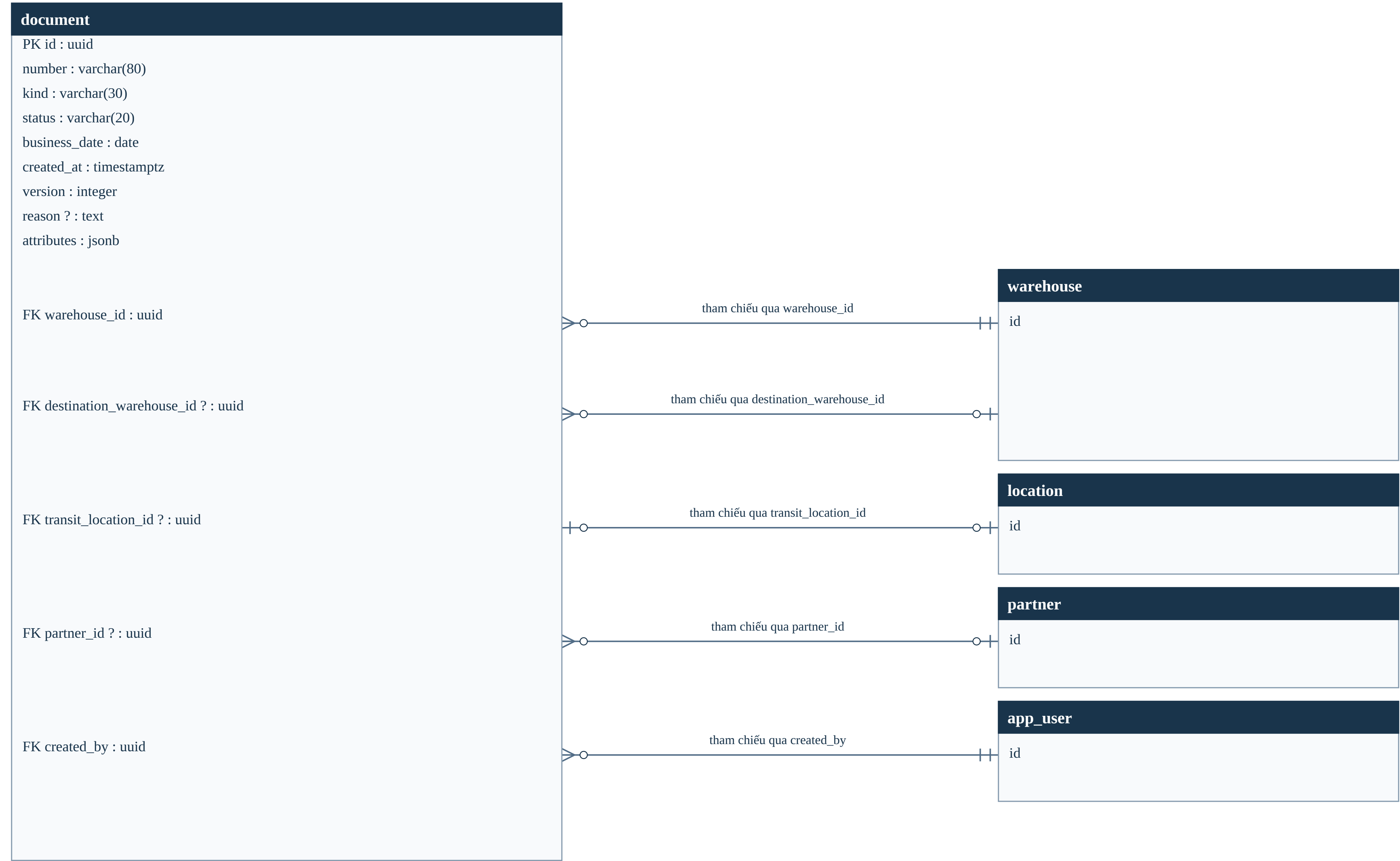
ERD vật lý / master / reference_price

WMS / BA 2.0 / 27-09-2026 / DRAFT



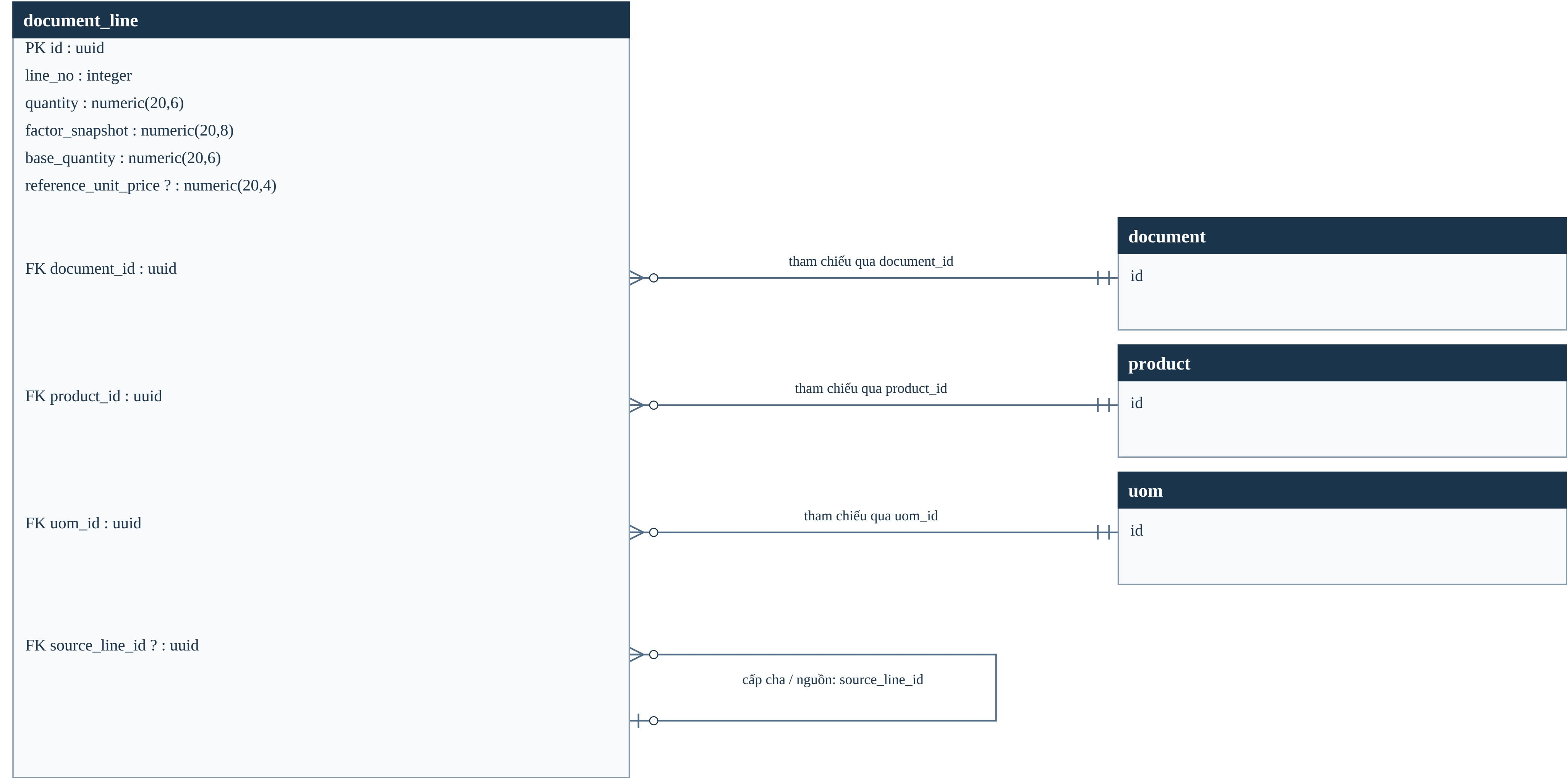
ERD vật lý / documents / document

WMS / BA 2.0 / 27-09-2026 / DRAFT



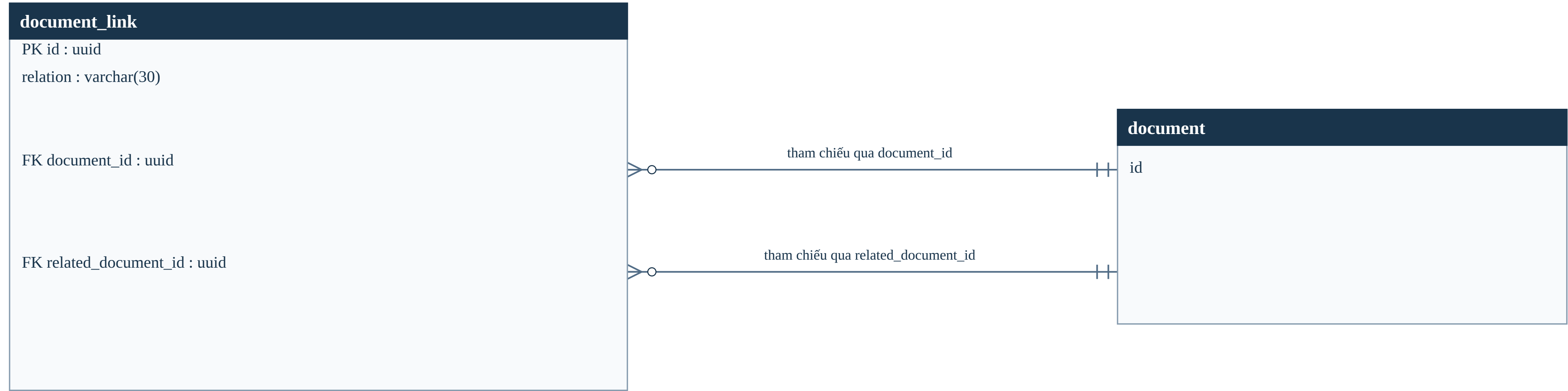
ERD vật lý / documents / document_line

WMS / BA 2.0 / 27-09-2026 / DRAFT



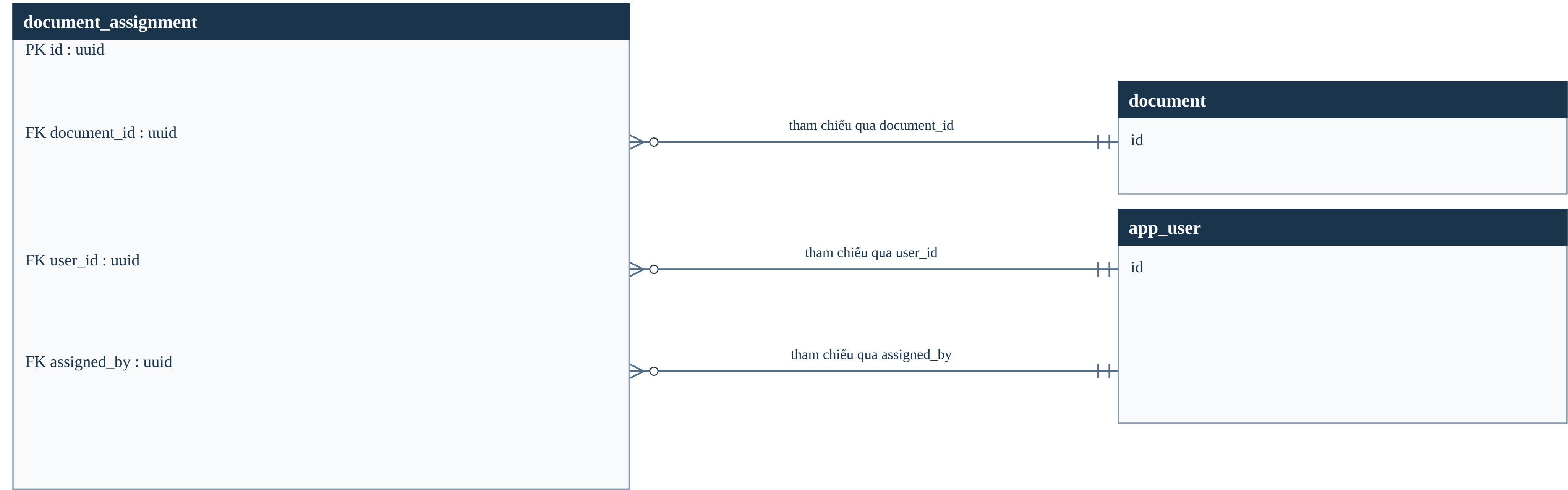
ERD vật lý / documents / document_link

WMS / BA 2.0 / 27-09-2026 / DRAFT



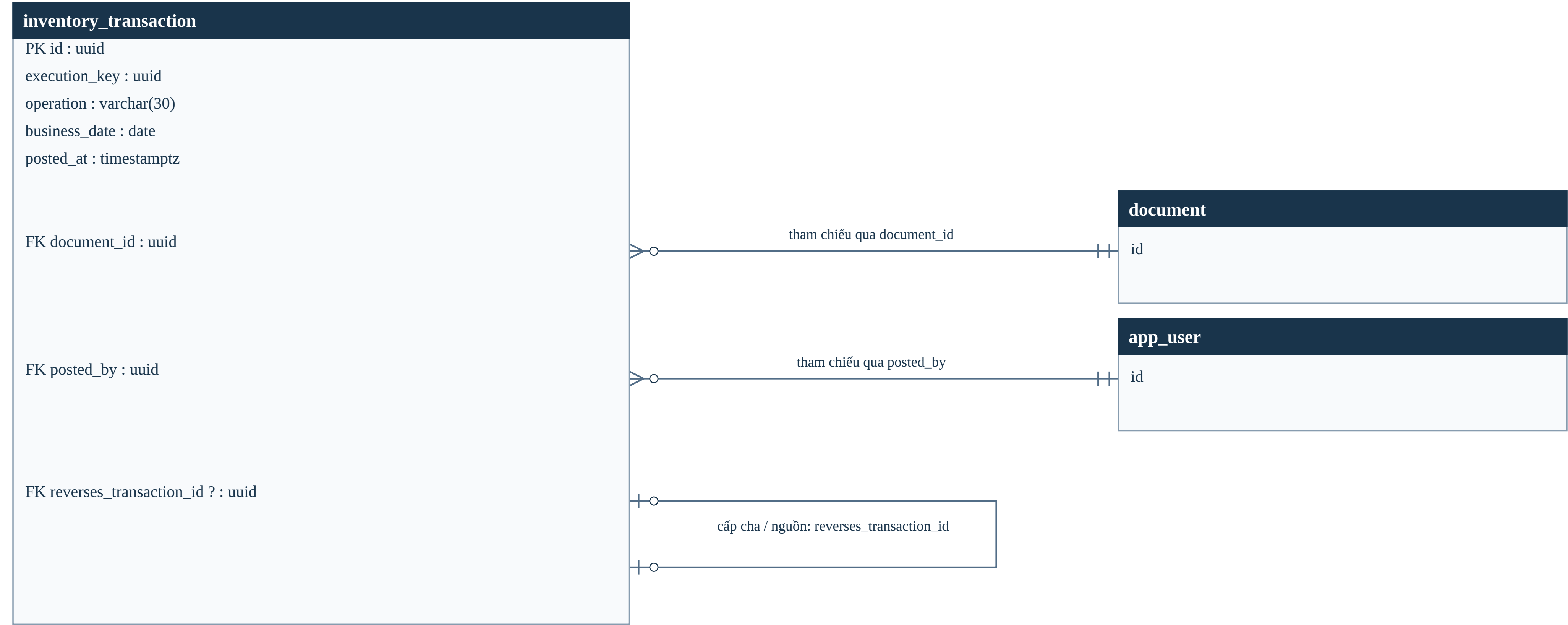
ERD vật lý / documents / document_assignment

WMS / BA 2.0 / 27-09-2026 / DRAFT



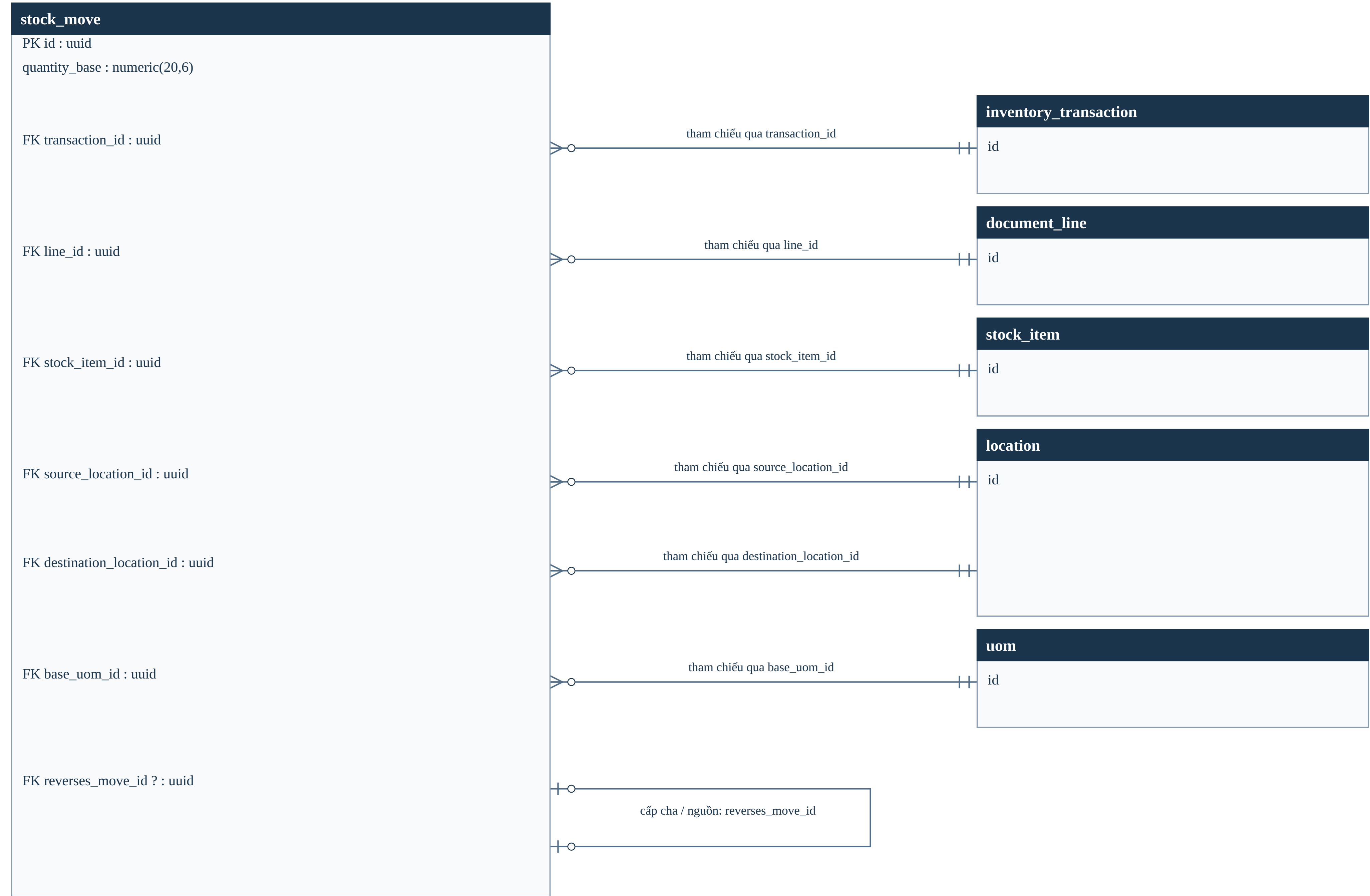
ERD vật lý / inventory / inventory_transaction

WMS / BA 2.0 / 27-09-2026 / DRAFT



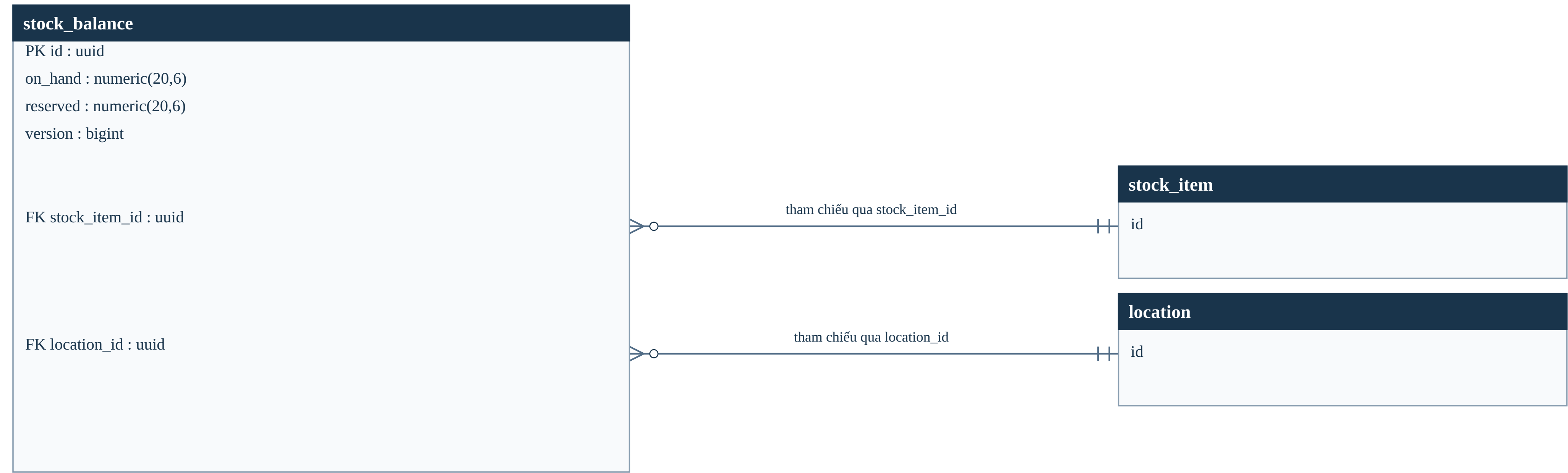
ERD vật lý / inventory / stock_move

WMS / BA 2.0 / 27-09-2026 / DRAFT



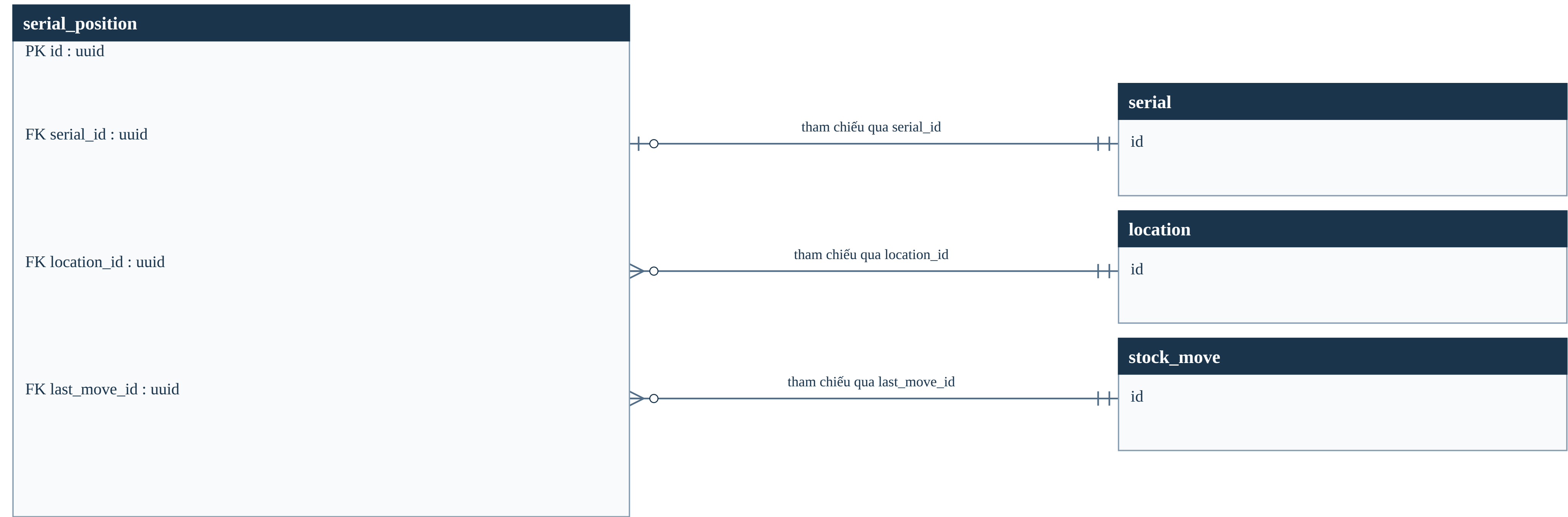
ERD vật lý / inventory / stock_balance

WMS / BA 2.0 / 27-09-2026 / DRAFT



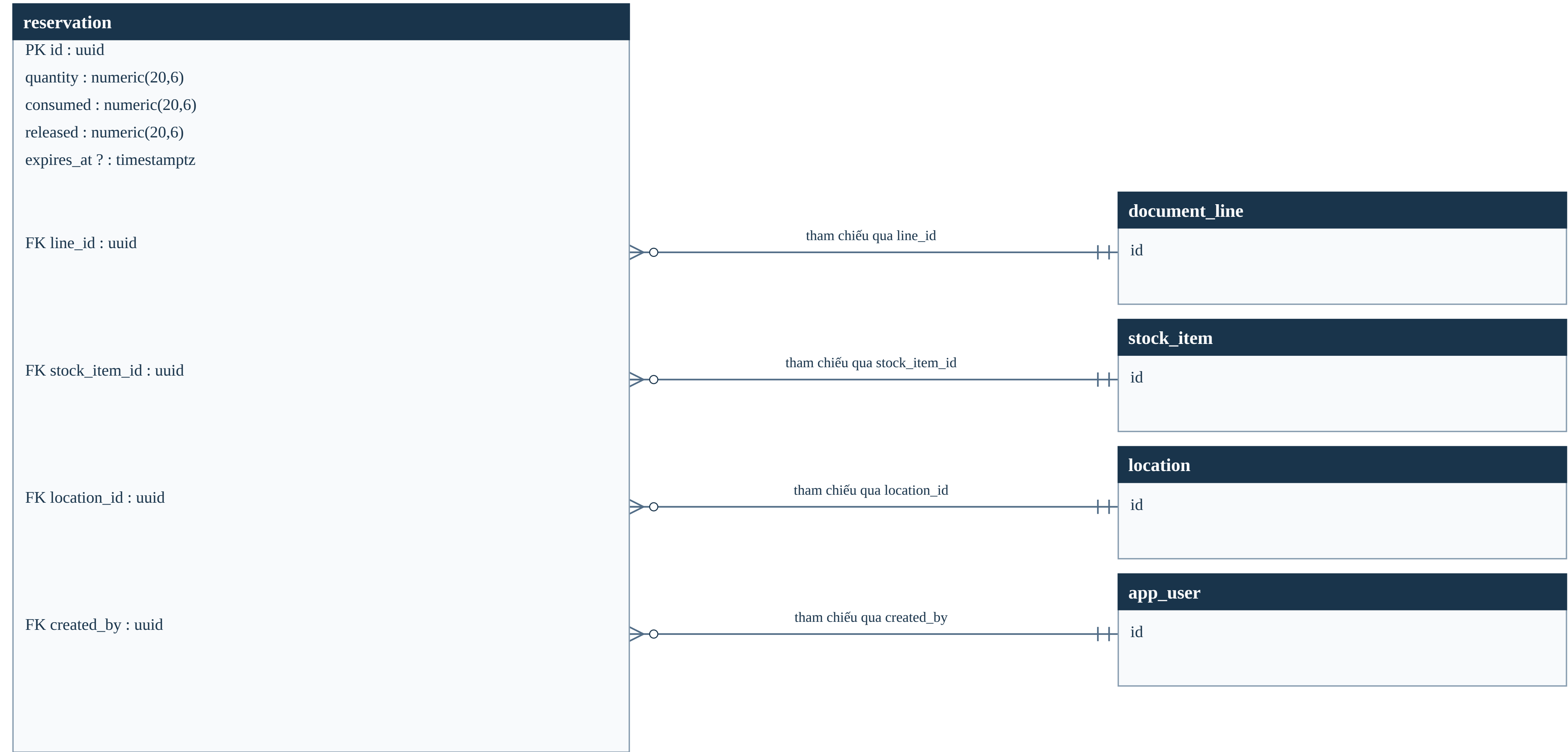
ERD vật lý / inventory / serial_position

WMS / BA 2.0 / 27-09-2026 / DRAFT



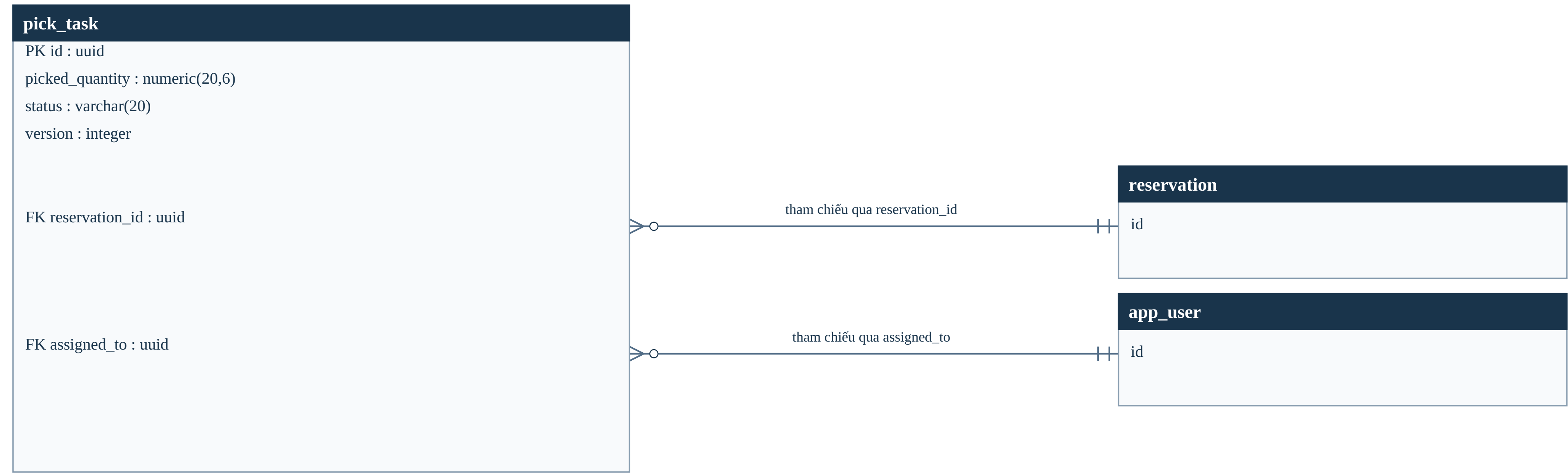
ERD vật lý / fulfillment / reservation

WMS / BA 2.0 / 27-09-2026 / DRAFT



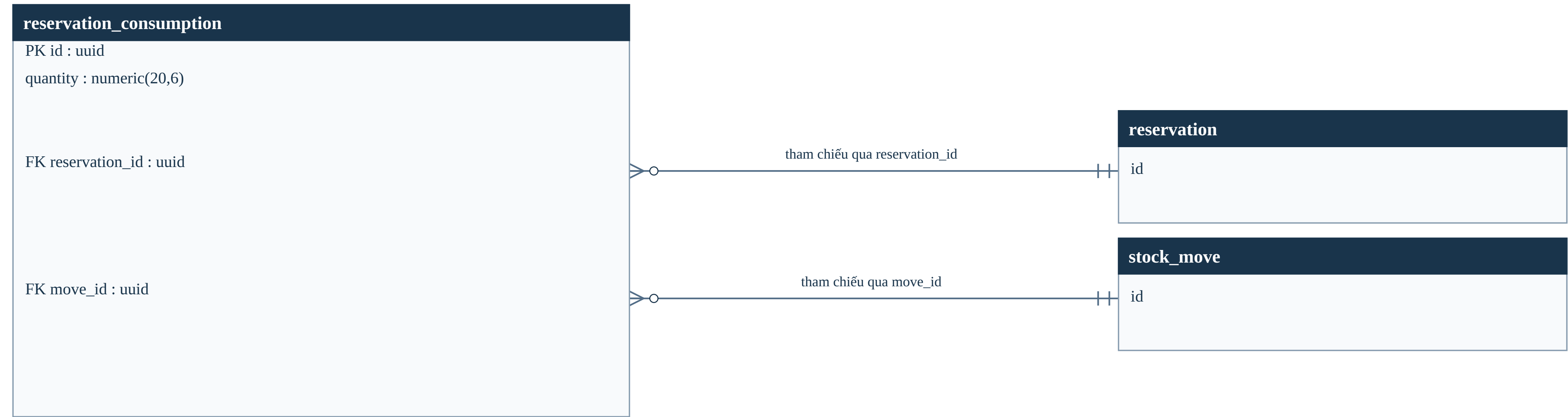
ERD vật lý / fulfillment / pick_task

WMS / BA 2.0 / 27-09-2026 / DRAFT



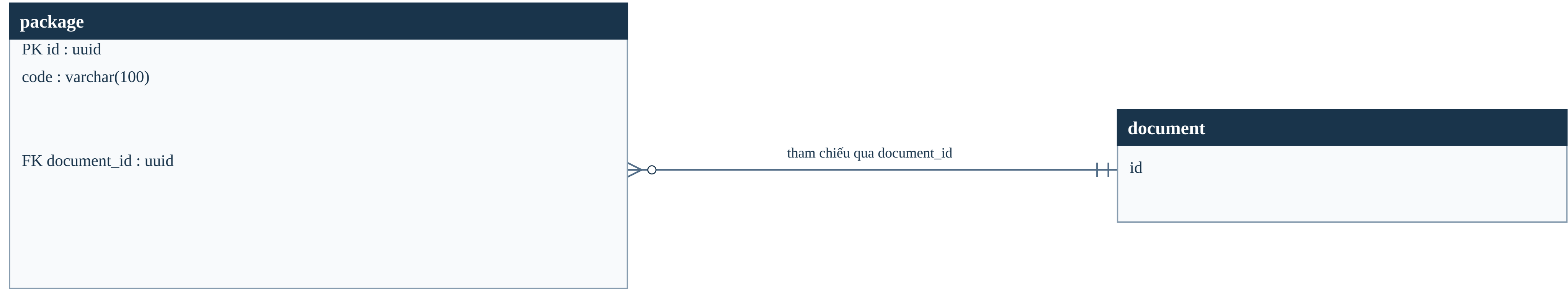
ERD vật lý / fulfillment / reservation_consumption

WMS / BA 2.0 / 27-09-2026 / DRAFT



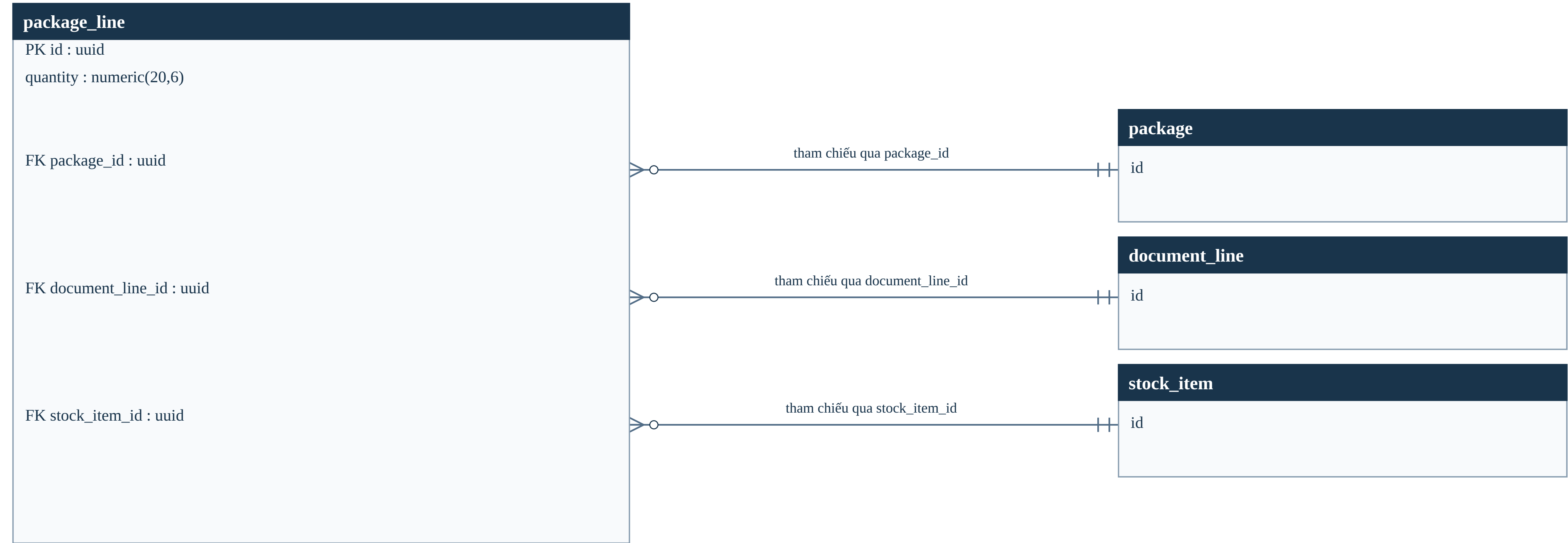
ERD vật lý / fulfillment / package

WMS / BA 2.0 / 27-09-2026 / DRAFT



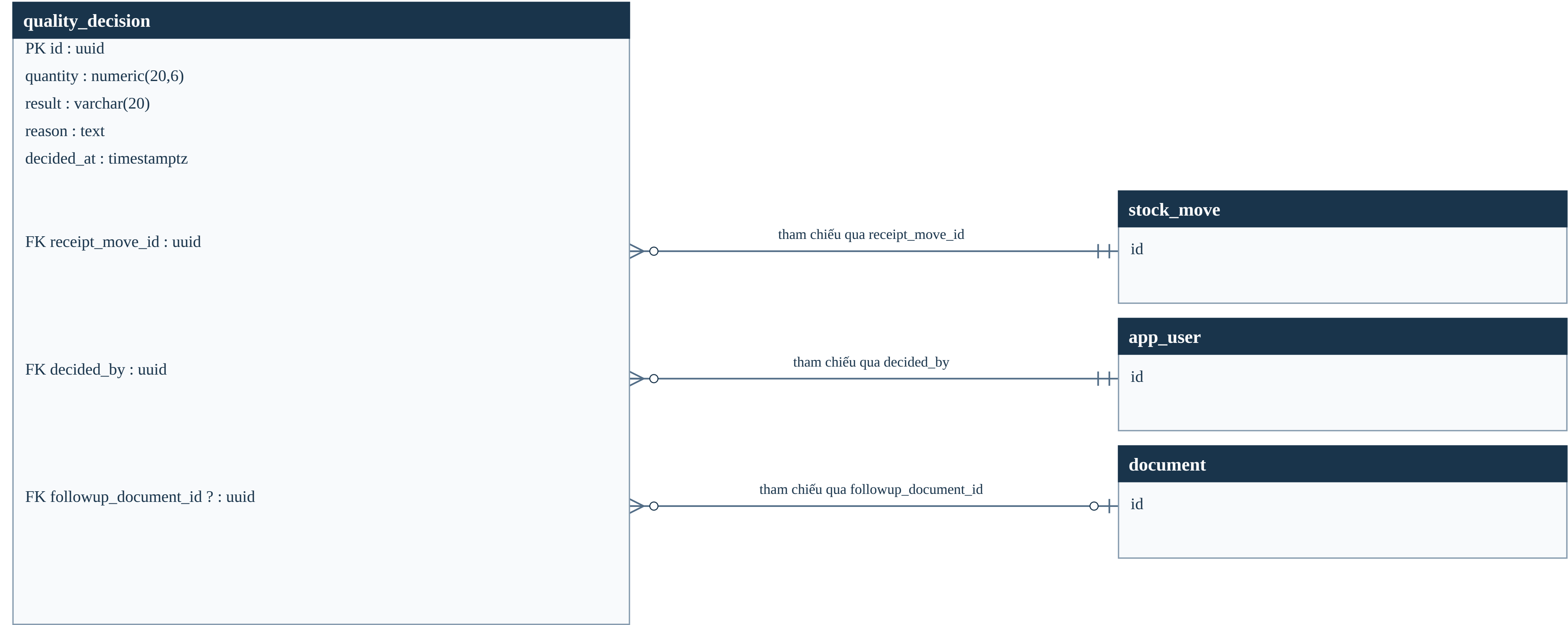
ERD vật lý / fulfillment / package_line

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ERD vật lý / fulfillment / quality_decision

WMS / BA 2.0 / 27-09-2026 / DRAFT



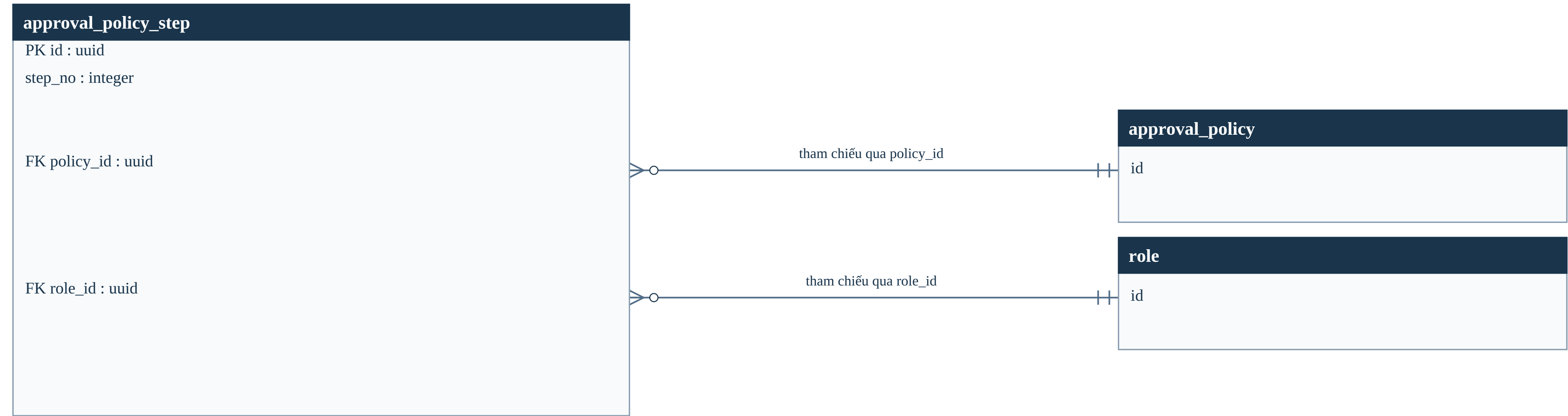
ERD vật lý / approval / approval_policy

WMS / BA 2.0 / 27-09-2026 / DRAFT

approval_policy
PK id : uuid
document_kind : varchar(30)
revision : integer
is_active : boolean

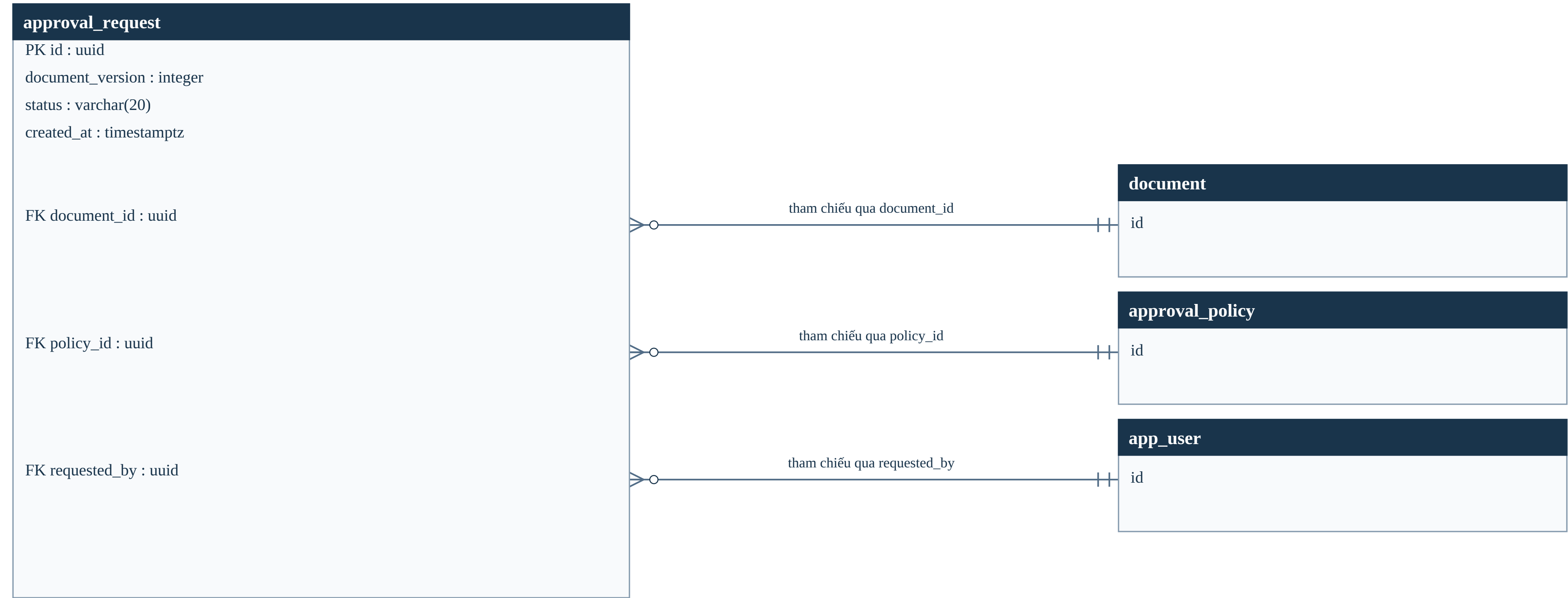
ERD vật lý / approval / approval_policy_step

WMS / BA 2.0 / 27-09-2026 / DRAFT



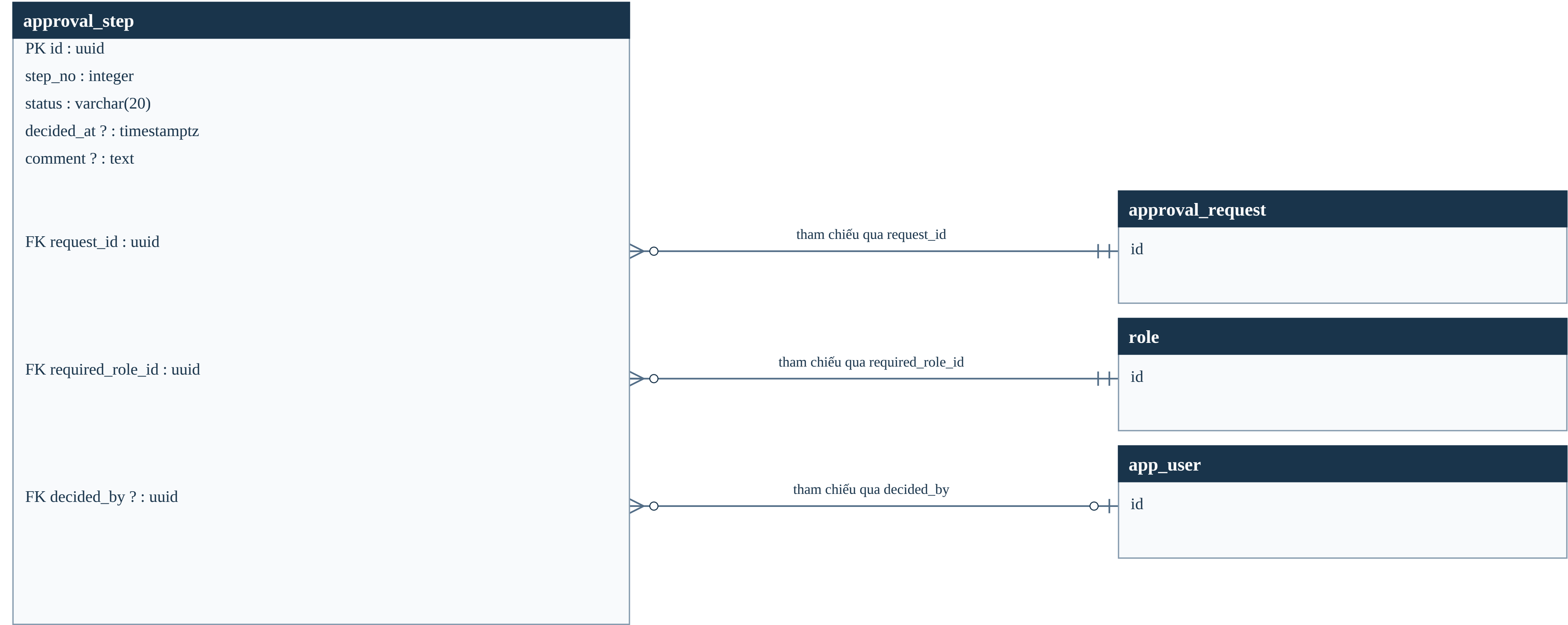
ERD vật lý / approval / approval_request

WMS / BA 2.0 / 27-09-2026 / DRAFT



ERD vật lý / approval / approval_step

WMS / BA 2.0 / 27-09-2026 / DRAFT



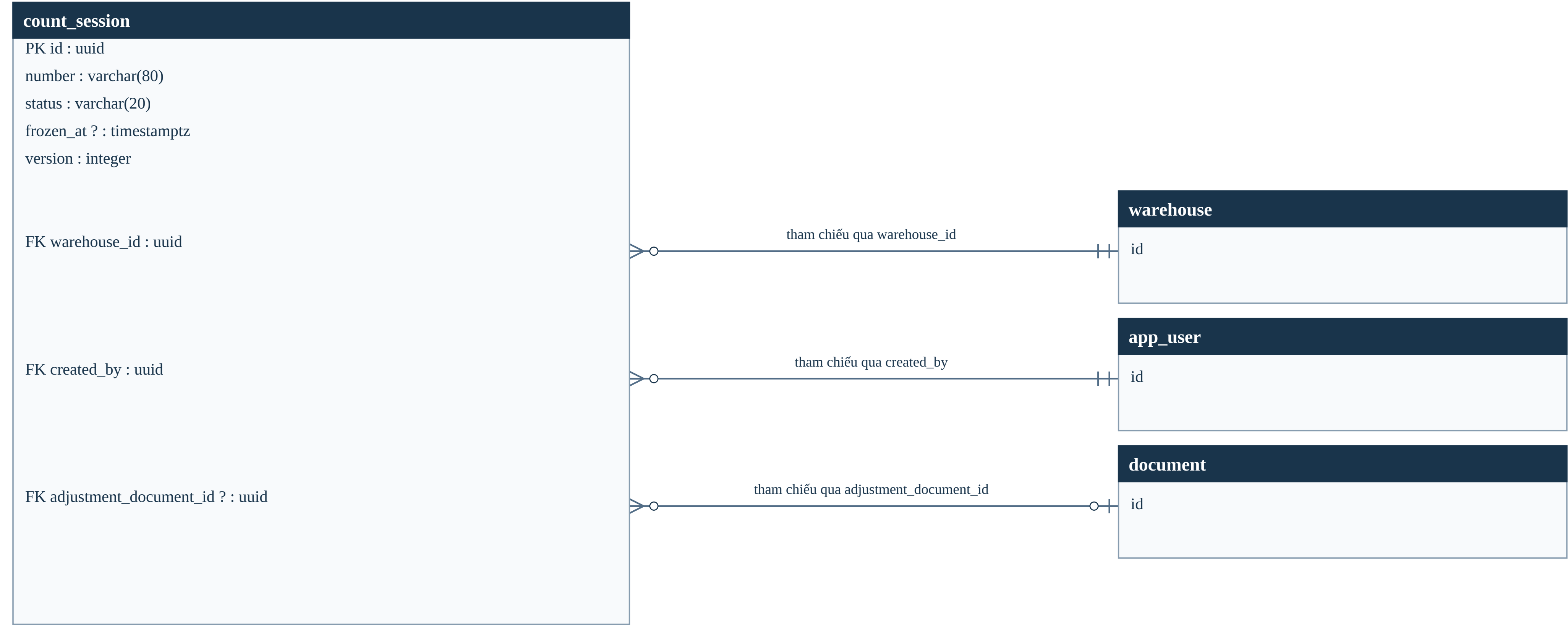
ERD vật lý / counting / stock_period

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ERD vật lý / counting / count_session

WMS / BA 2.0 / 27-09-2026 / DRAFT



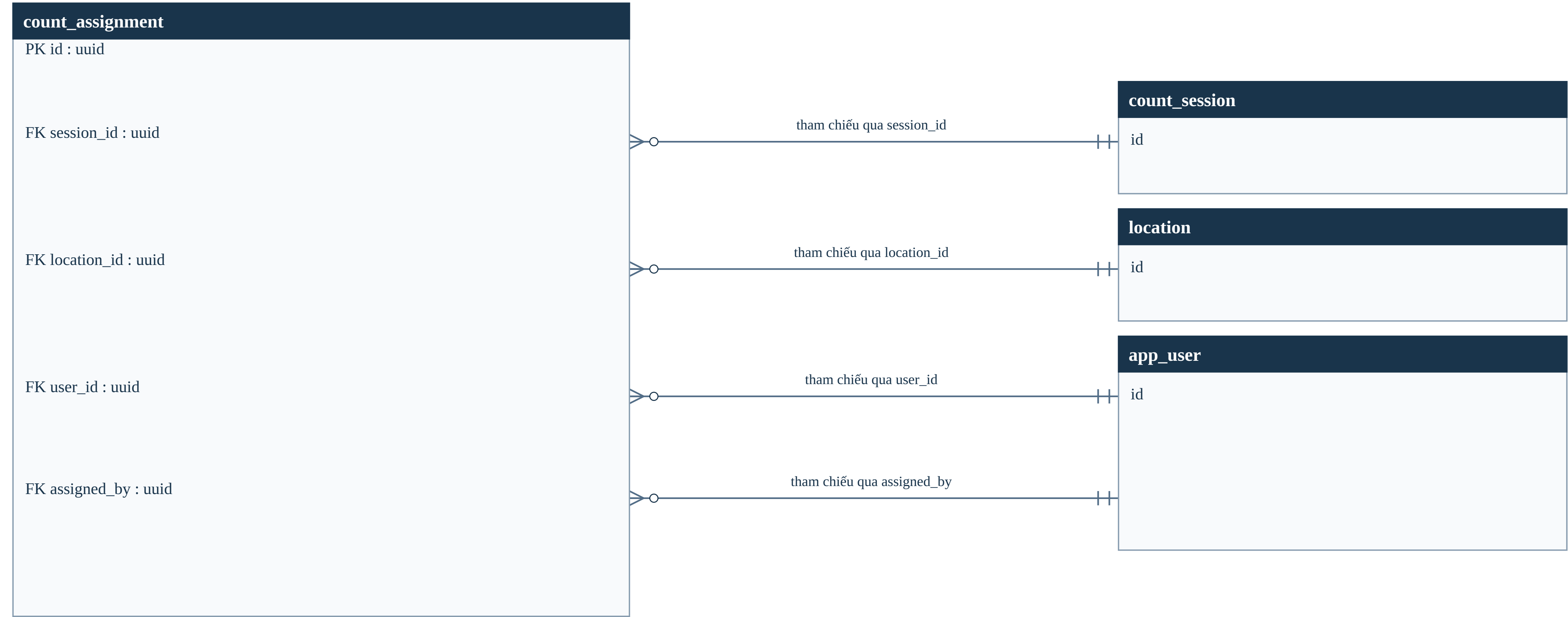
ERD vật lý / counting / count_location_lock

WMS / BA 2.0 / 27-09-2026 / DRAFT



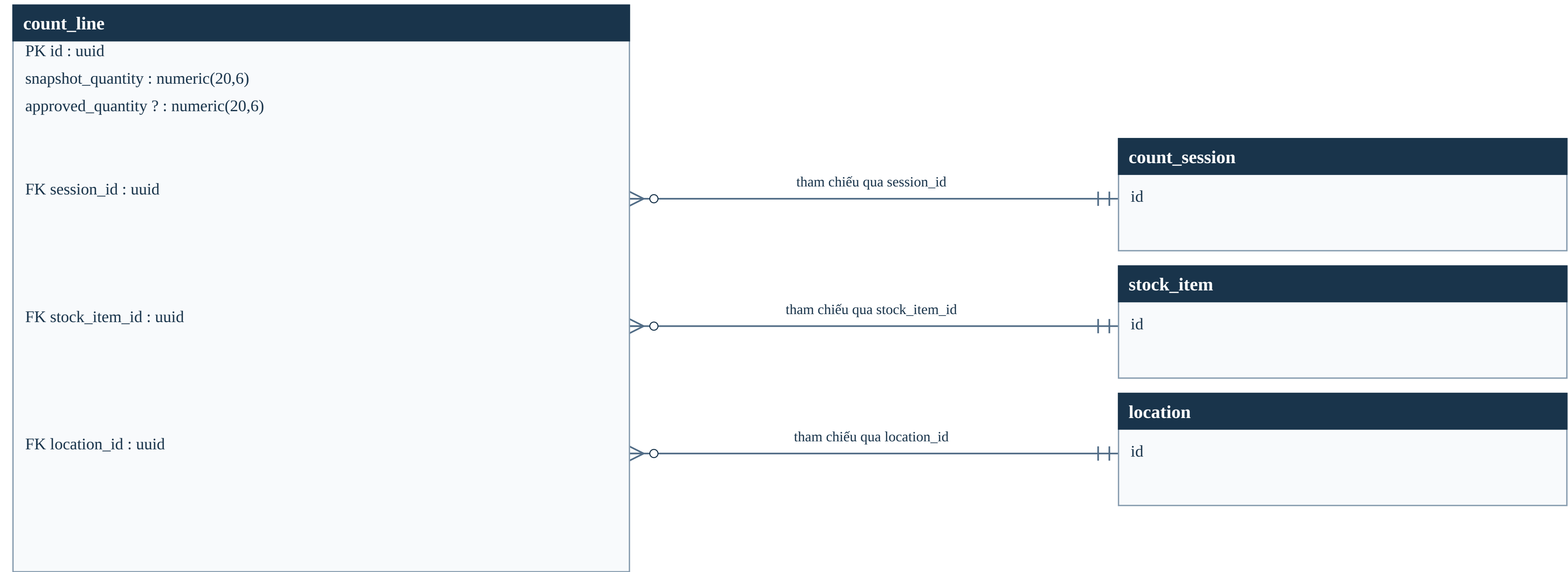
ERD vật lý / counting / count_assignment

WMS / BA 2.0 / 27-09-2026 / DRAFT



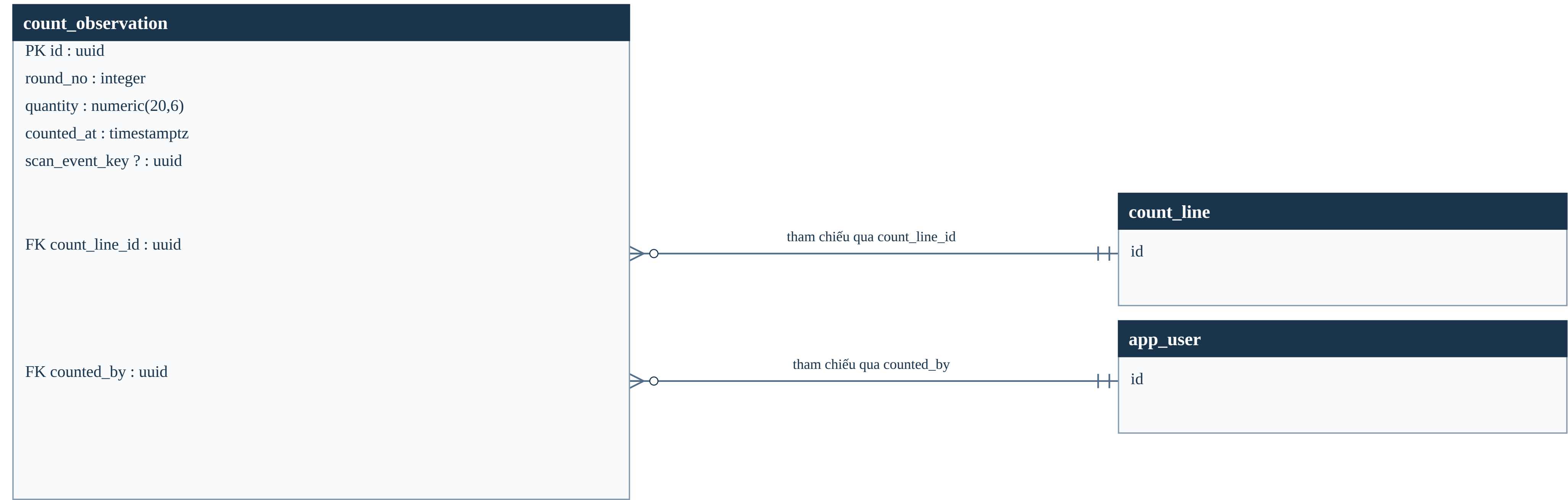
ERD vật lý / counting / count_line

WMS / BA 2.0 / 27-09-2026 / DRAFT



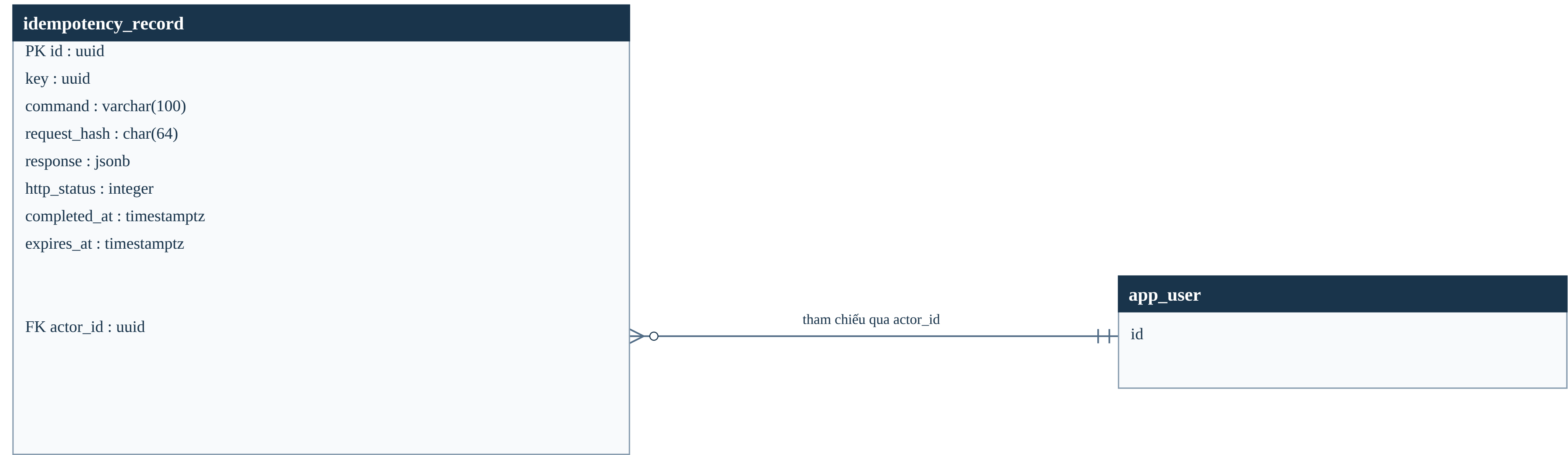
ERD vật lý / counting / count_observation

WMS / BA 2.0 / 27-09-2026 / DRAFT



ERD vật lý / operations / idempotency_record

WMS / BA 2.0 / 27-09-2026 / DRAFT



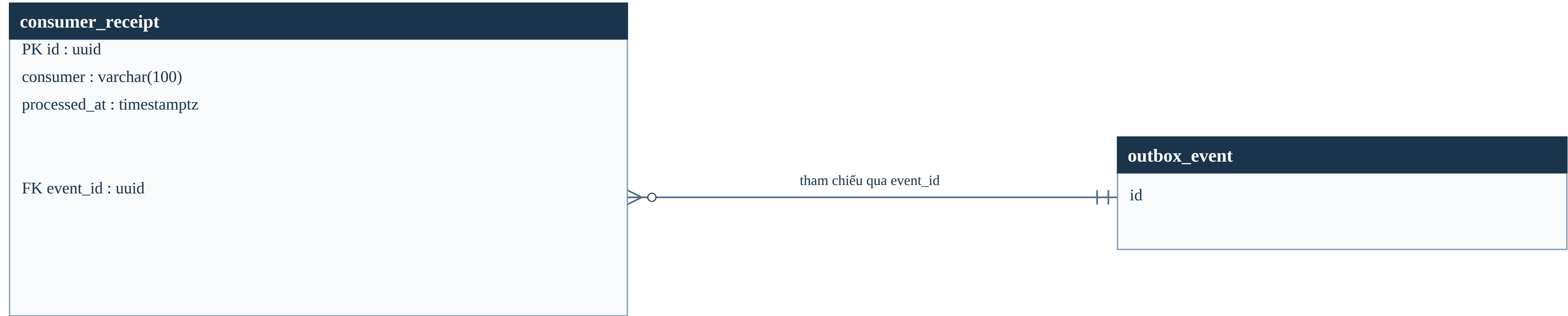
ERD vật lý / operations / outbox_event

WMS / BA 2.0 / 27-09-2026 / DRAFT

outbox_event
PK id : uuid
event_type : varchar(100)
aggregate_id : uuid
payload : jsonb
occurred_at : timestamptz
available_at : timestamptz
attempts : integer
processed_at ? : timestamptz
last_error ? : text

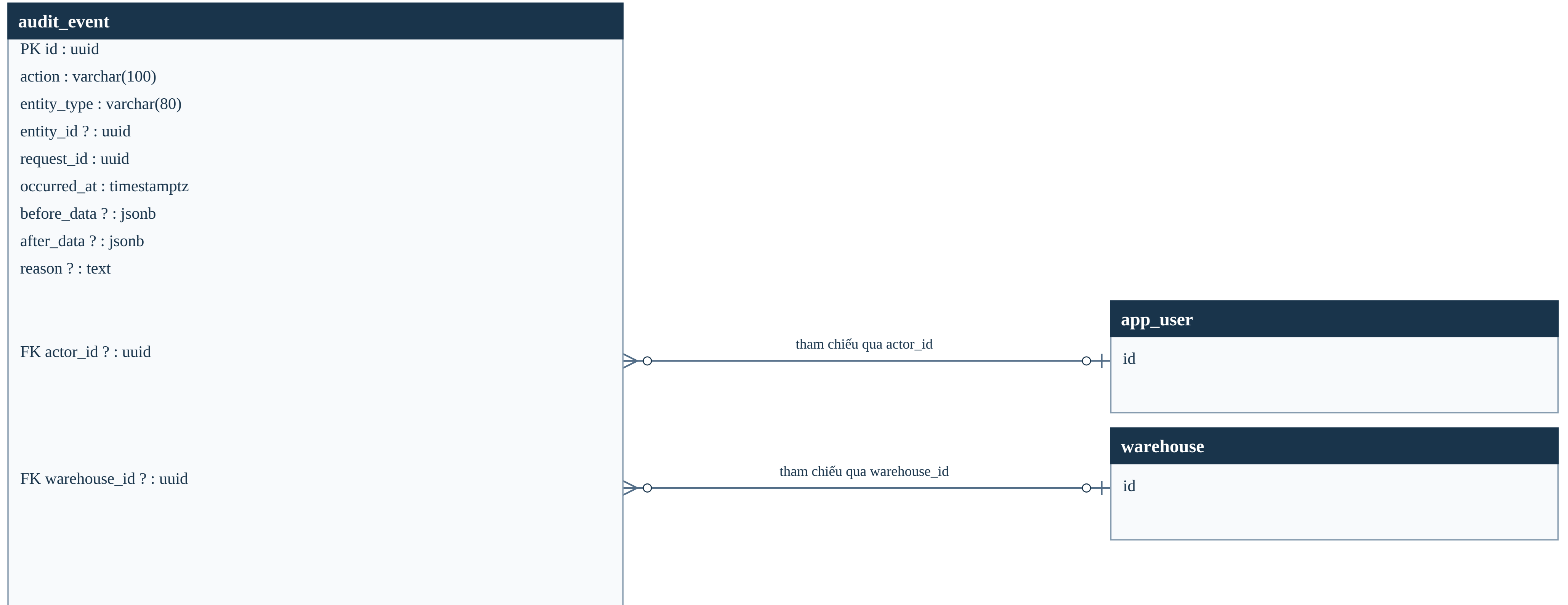
ERD vật lý / operations / consumer_receipt

WMS / BA 2.0 / 27-09-2026 / DRAFT



ERD vật lý / operations / audit_event

WMS / BA 2.0 / 27-09-2026 / DRAFT



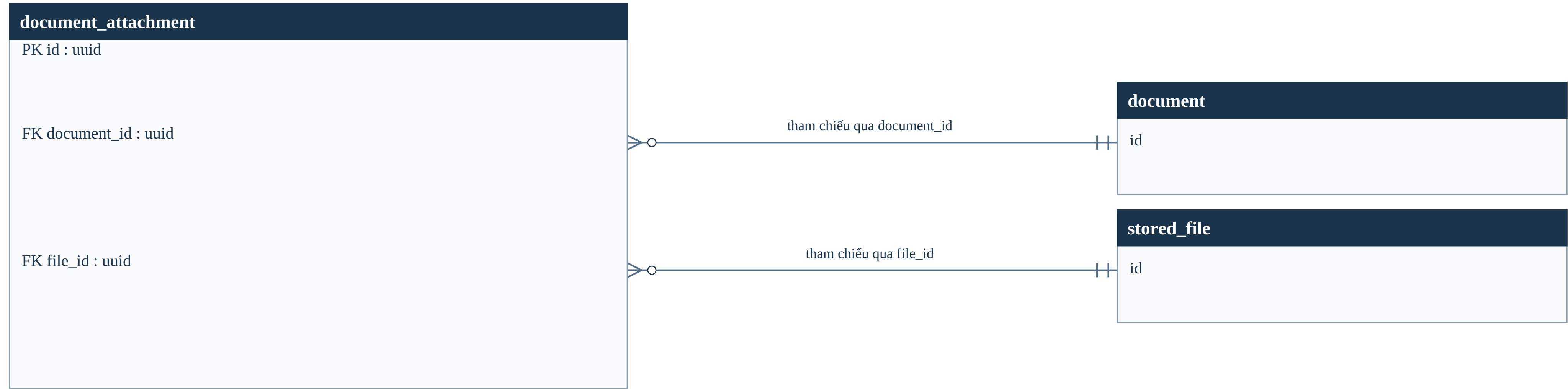
ERD vật lý / operations / stored_file

WMS / BA 2.0 / 27-09-2026 / DRAFT



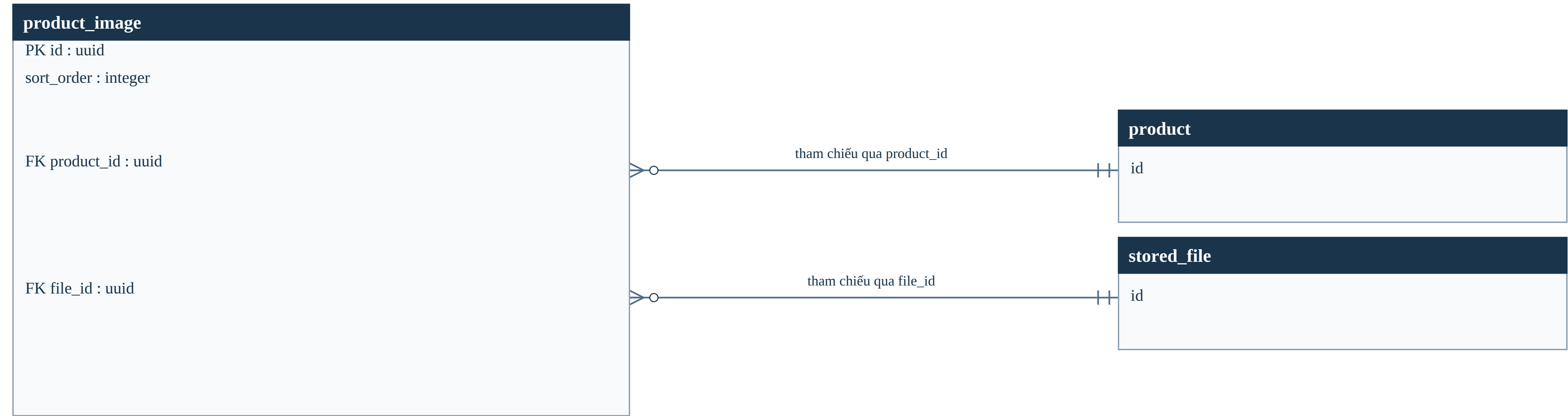
ERD vật lý / documents / document_attachment

WMS / BA 2.0 / 27-09-2026 / DRAFT



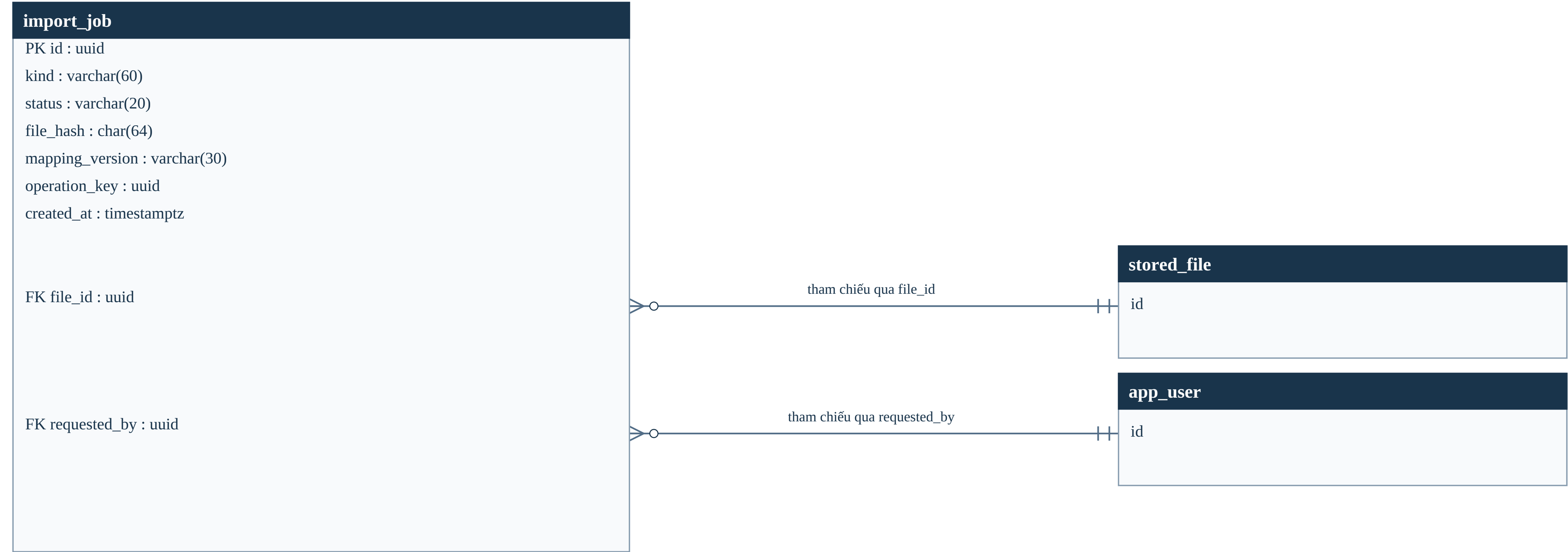
ERD vật lý / master / product_image

WMS / BA 2.0 / 27-09-2026 / DRAFT



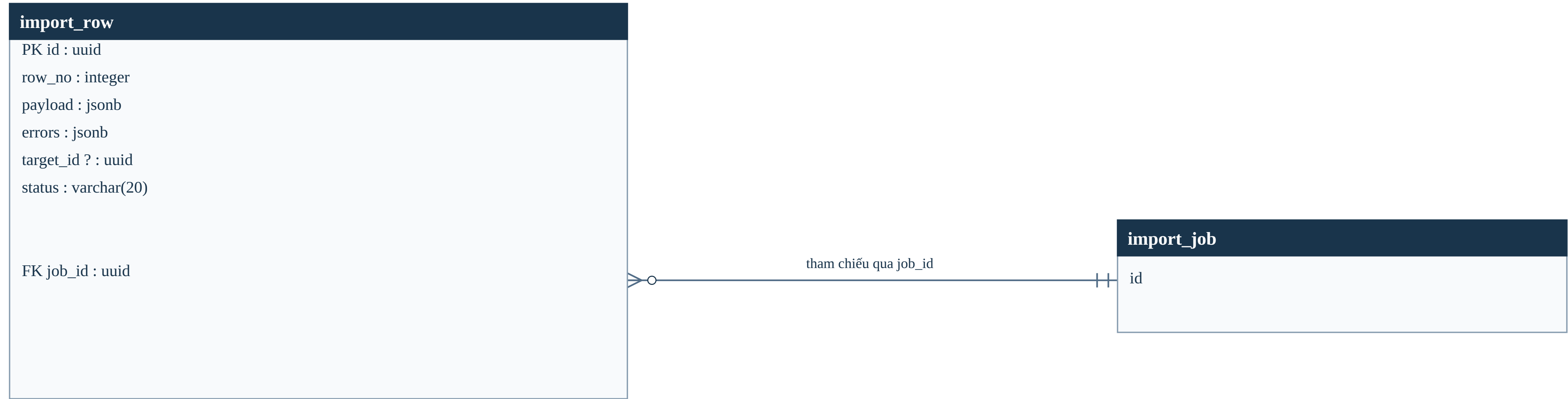
ERD vật lý / operations / import_job

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ERD vật lý / operations / import_row

WMS / BA 2.0 / 27-09-2026 / DRAFT



ERD vật lý / operations / export_job

WMS / BA 2.0 / 27-09-2026 / DRAFT



ERD vật lý / extension / custom_field_definition

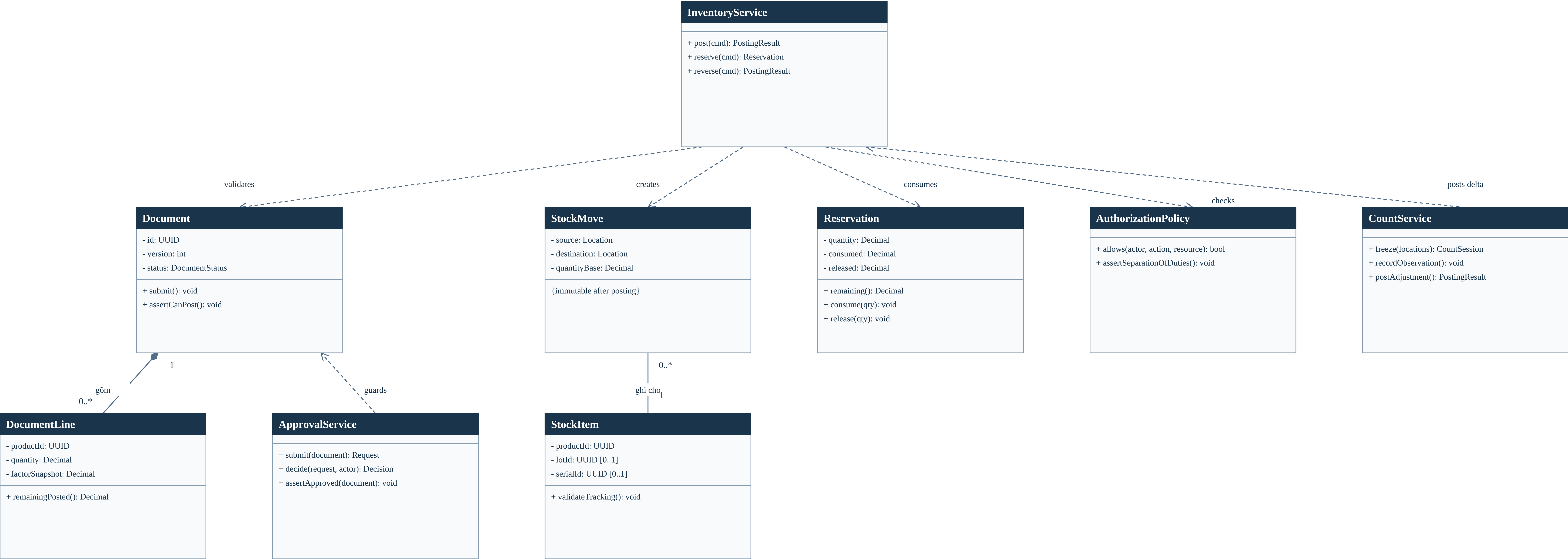
WMS / BA 2.0 / 27-09-2026 / DRAFT

custom_field_definition
PK id : uuid
entity_type : varchar(40)
code : varchar(60)
value_type : varchar(20)
required : boolean
validation : jsonb
is_active : boolean

ERD vật lý / extension / client_release

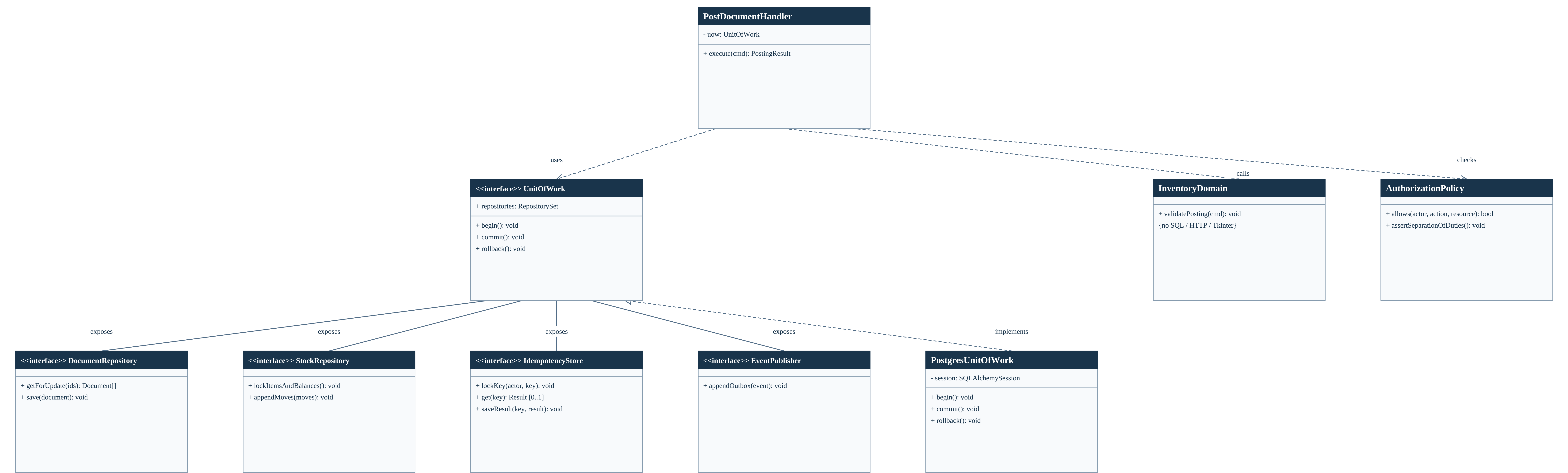
WMS / BA 2.0 / 27-09-2026 / DRAFT

client_release
PK id : uuid
version : varchar(40)
os : varchar(20)
architecture : varchar(20)
api_major : integer
sha256 : char(64)
download_uri : text
is_minimum : boolean



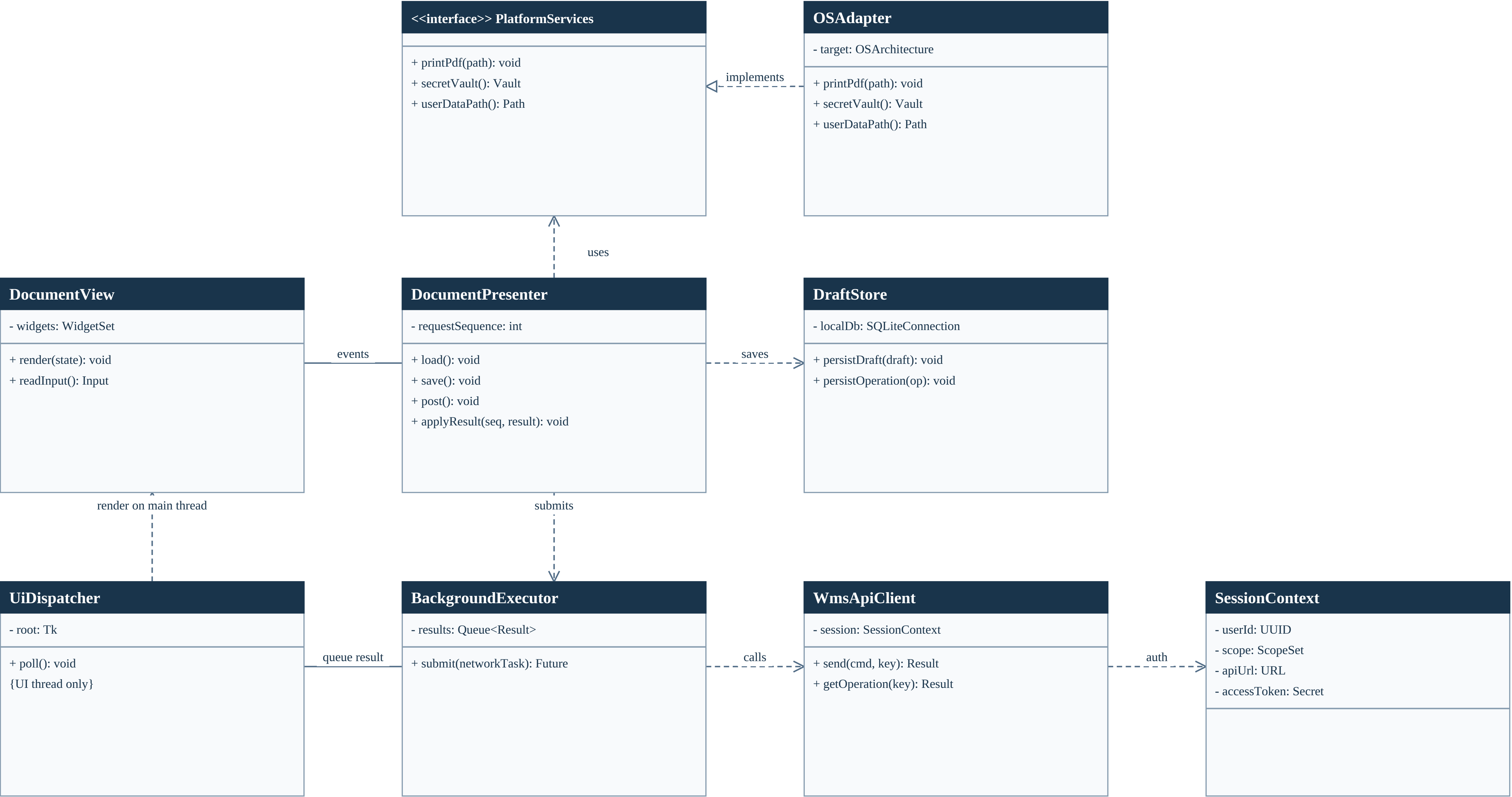
Class Application / UML implementation

WMS / BA 2.0 / 27-09-2026 / DRAFT



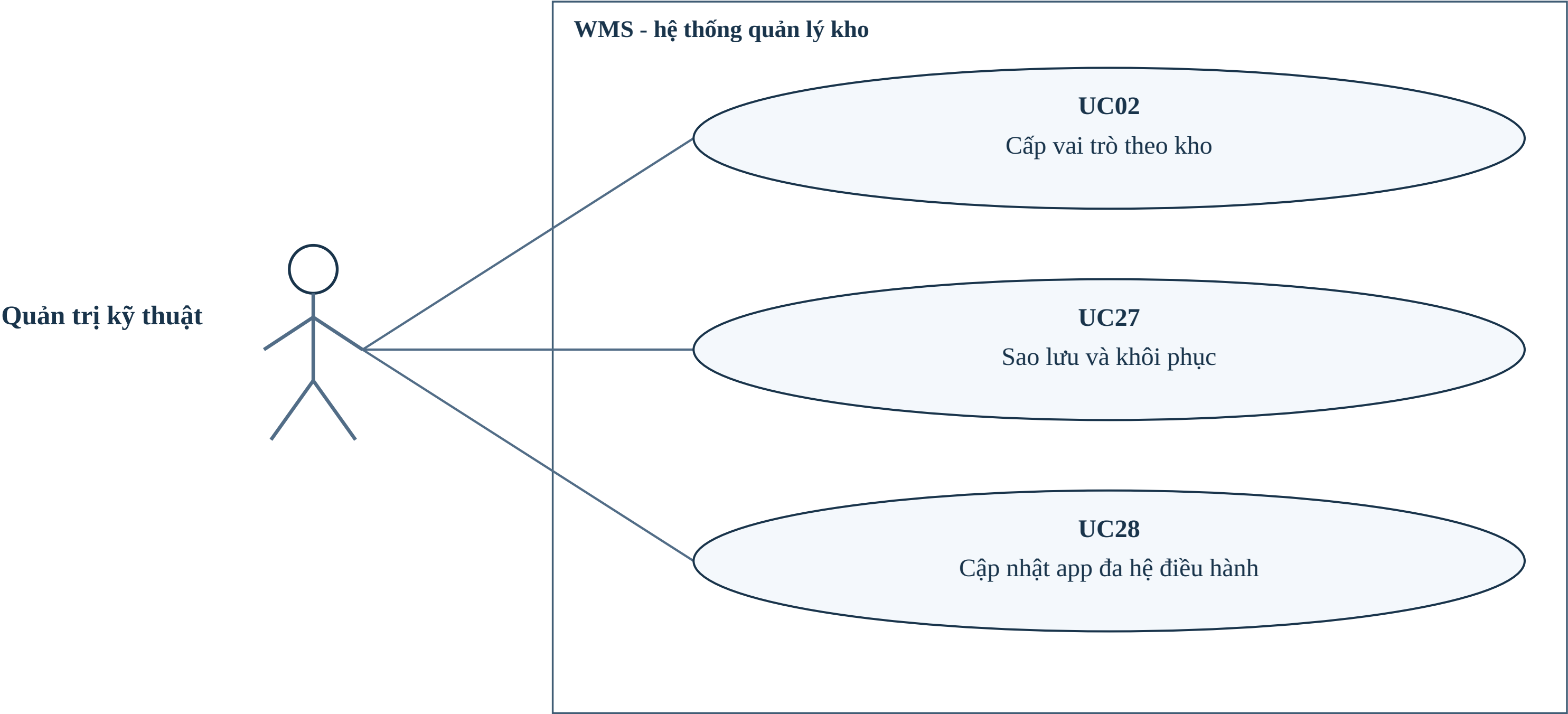
Class Desktop / UML implementation

WMS / BA 2.0 / 27-09-2026 / DRAFT



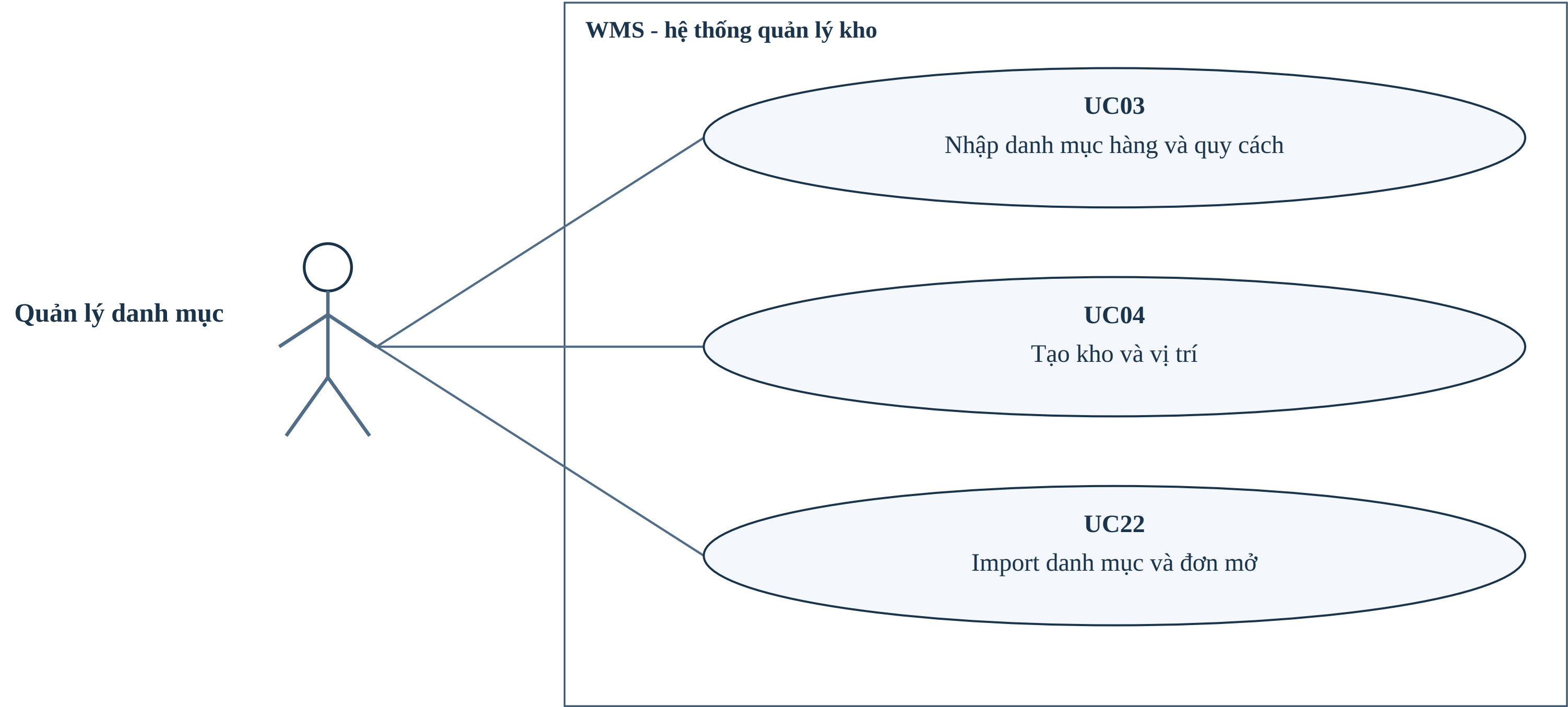
Quản trị kỹ thuật / Use case

WMS / BA 2.0 / 27-09-2026 / DRAFT



Quản lý danh mục / Use case

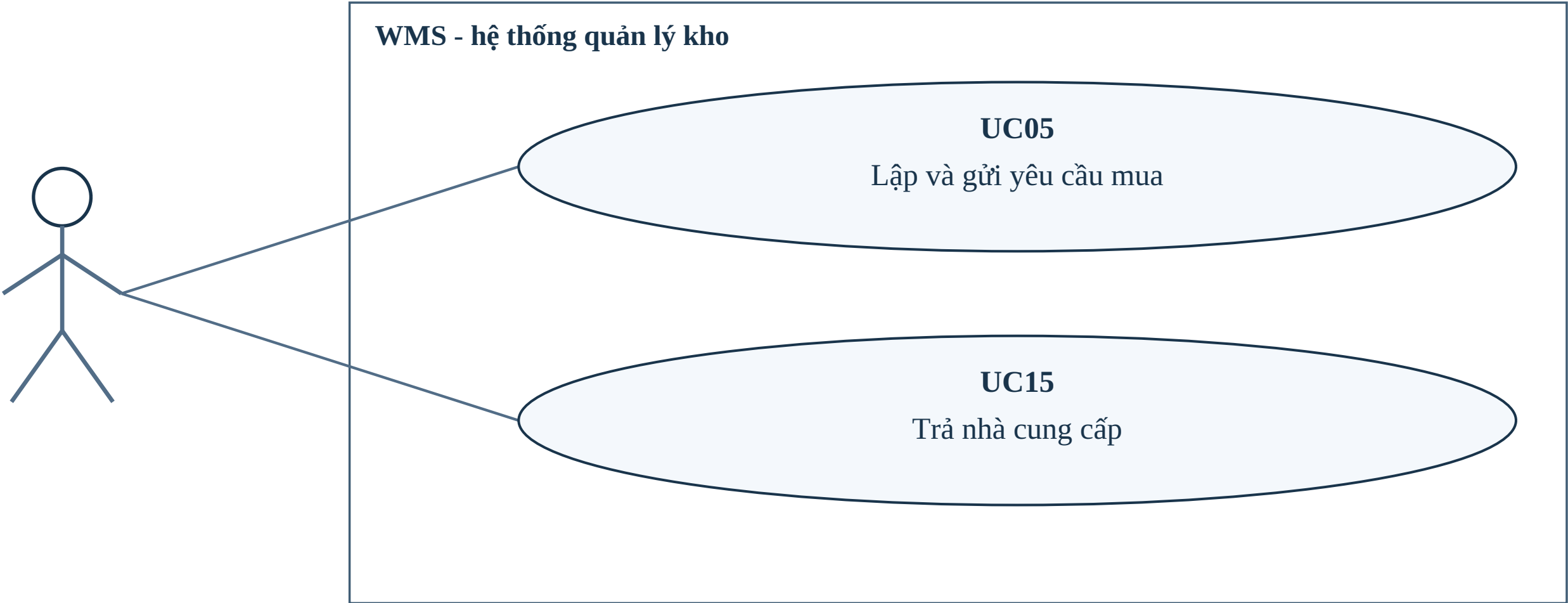
WMS / BA 2.0 / 27-09-2026 / DRAFT



Mua hàng / Use case

WMS / BA 2.0 / 27-09-2026 / DRAFT

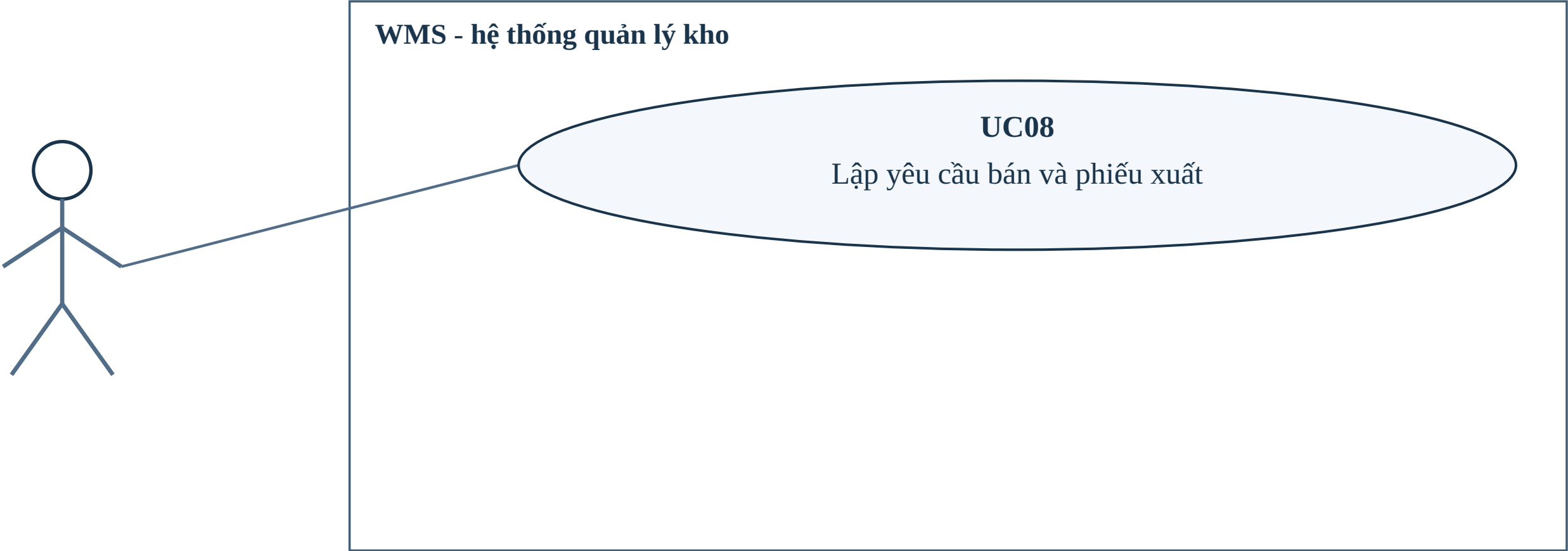
Mua hàng



Bán hàng / Use case

WMS / BA 2.0 / 27-09-2026 / DRAFT

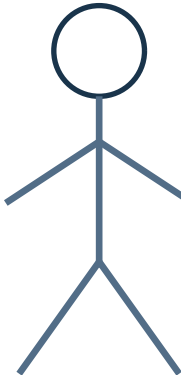
Bán hàng



Nhân viên nhận / Use case

WMS / BA 2.0 / 27-09-2026 / DRAFT

Nhân viên nhận



WMS - hệ thống quản lý kho

UC06

Nhận hàng từng phần

UC07

Kiểm tra và cất hàng

UC13

Nhận chuyển thiếu hoặc từng phần

UC14

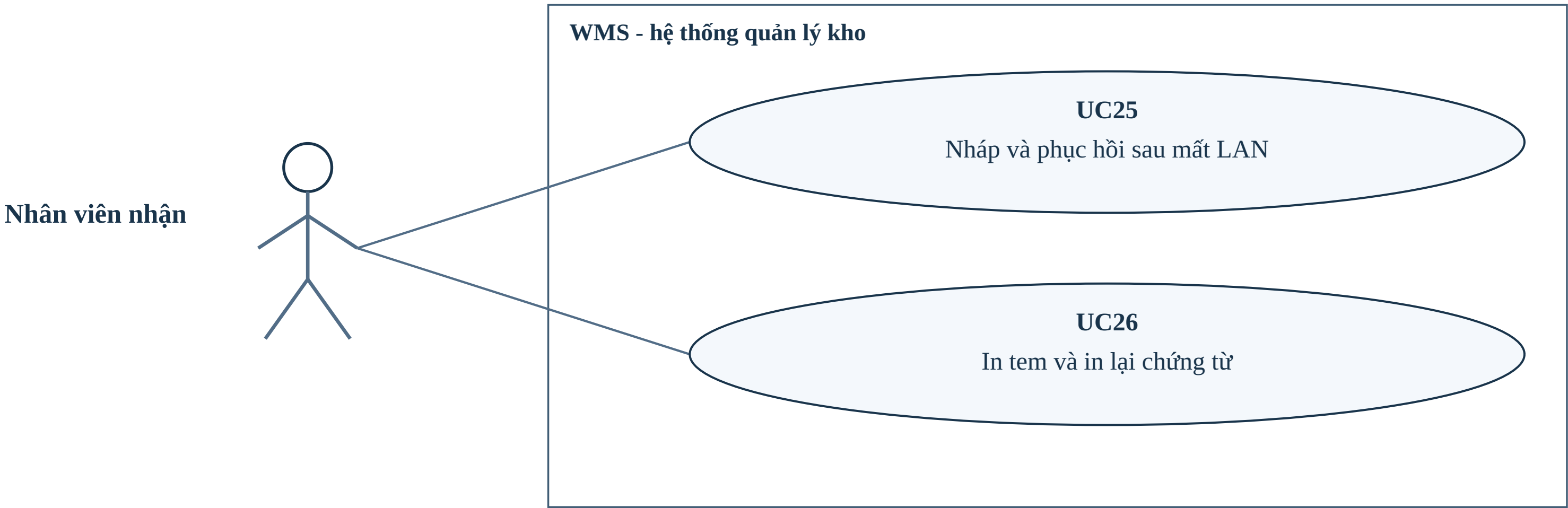
Khách trả hàng

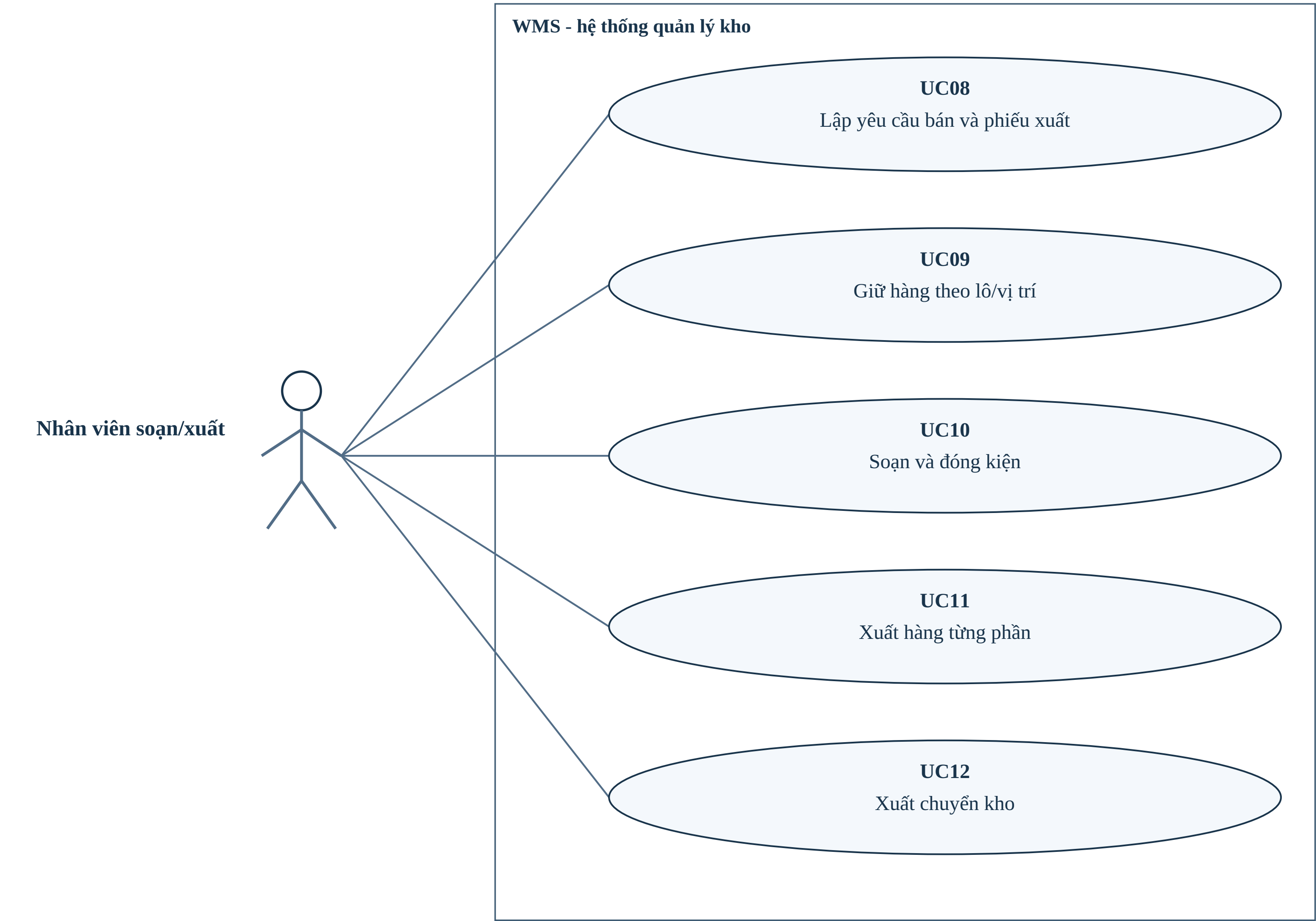
UC17

Đếm mù và đếm lại

Nhân viên nhận / Use case

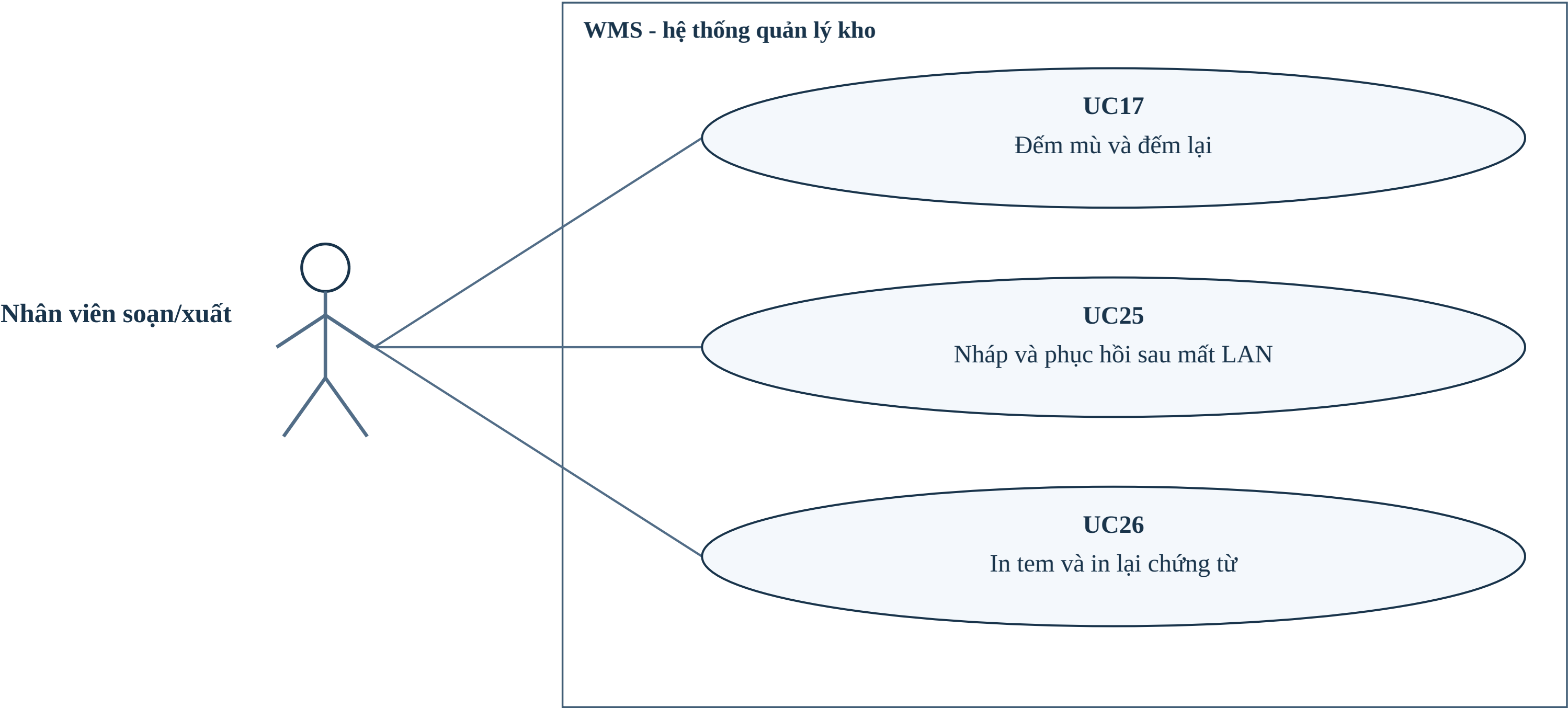
WMS / BA 2.0 / 27-09-2026 / DRAFT

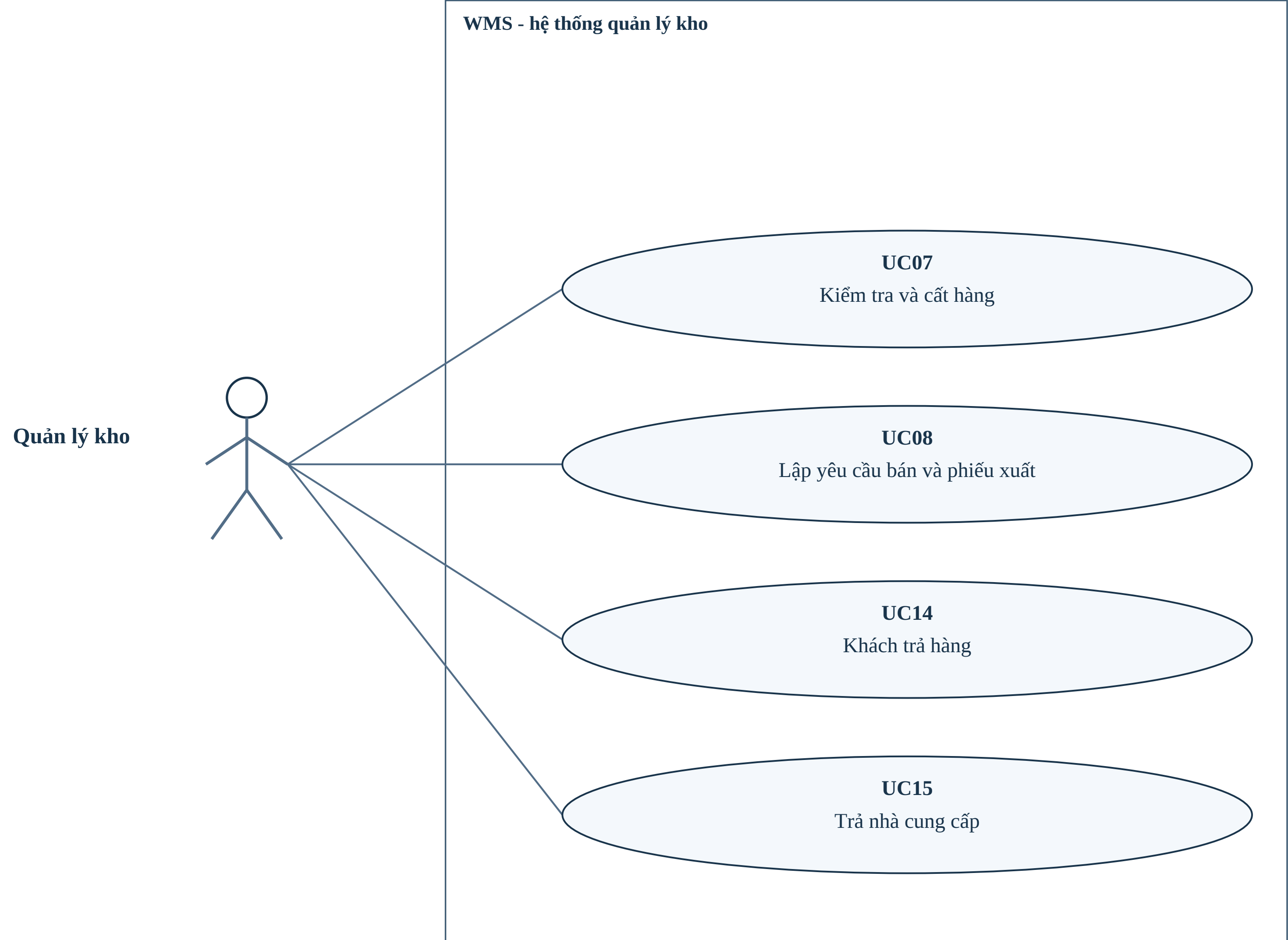


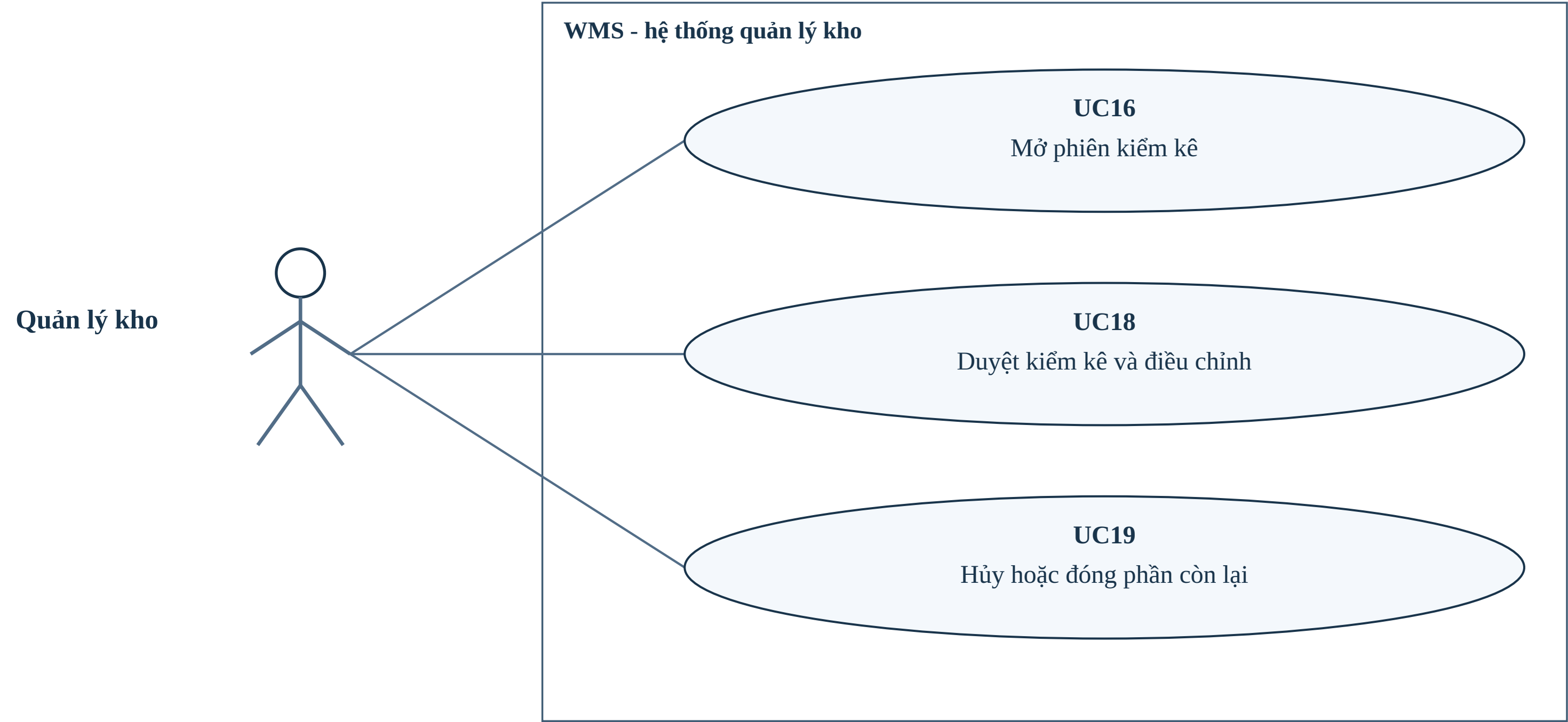


Nhân viên soạn/xuất / Use case

WMS / BA 2.0 / 27-09-2026 / DRAFT

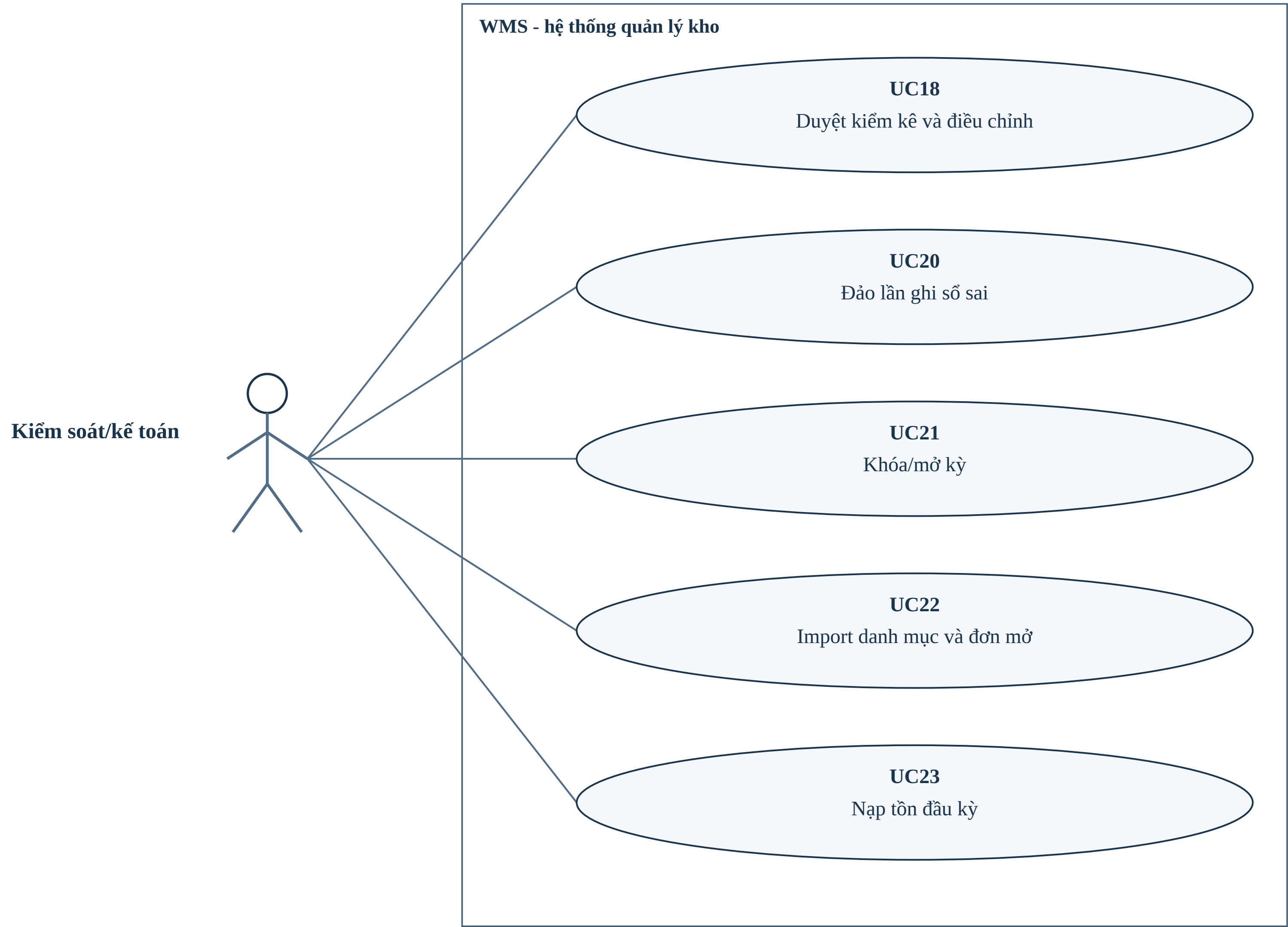






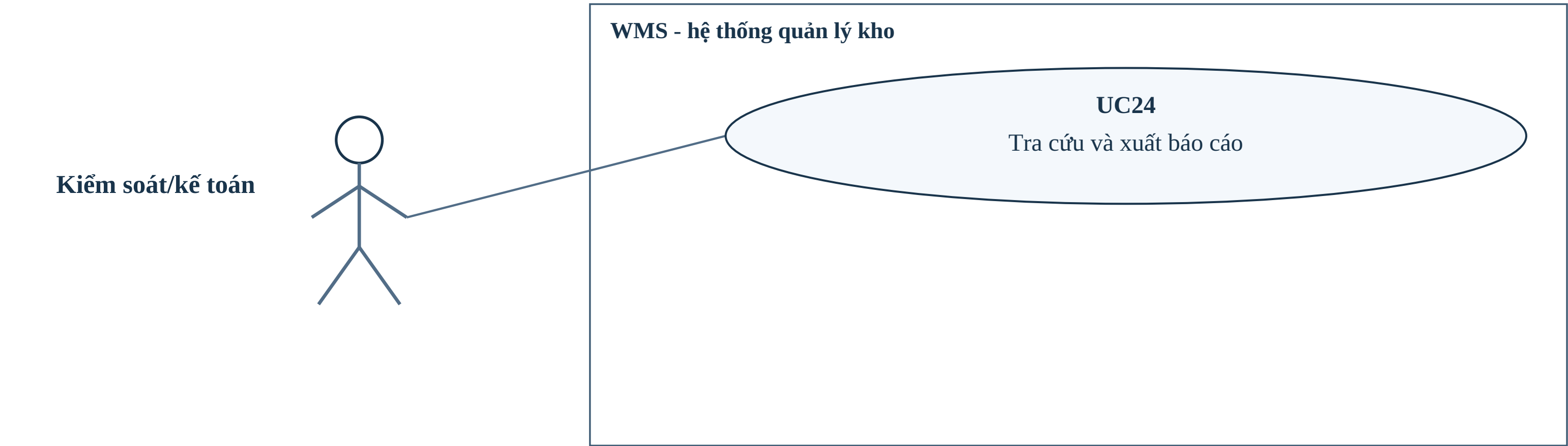
Kiểm soát/kế toán / Use case

WMS / BA 2.0 / 27-09-2026 / DRAFT



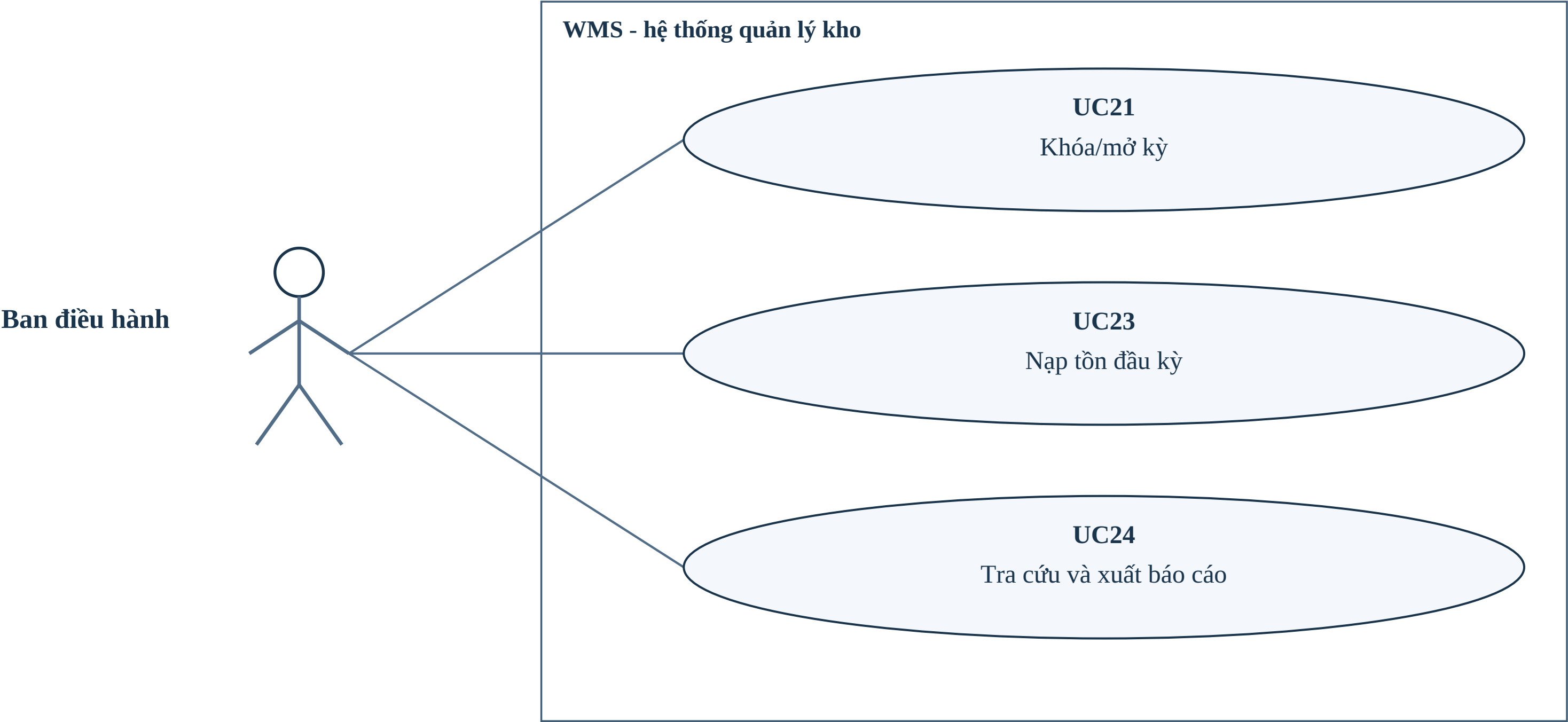
Kiểm soát/kế toán / Use case

WMS / BA 2.0 / 27-09-2026 / DRAFT



Ban điều hành / Use case

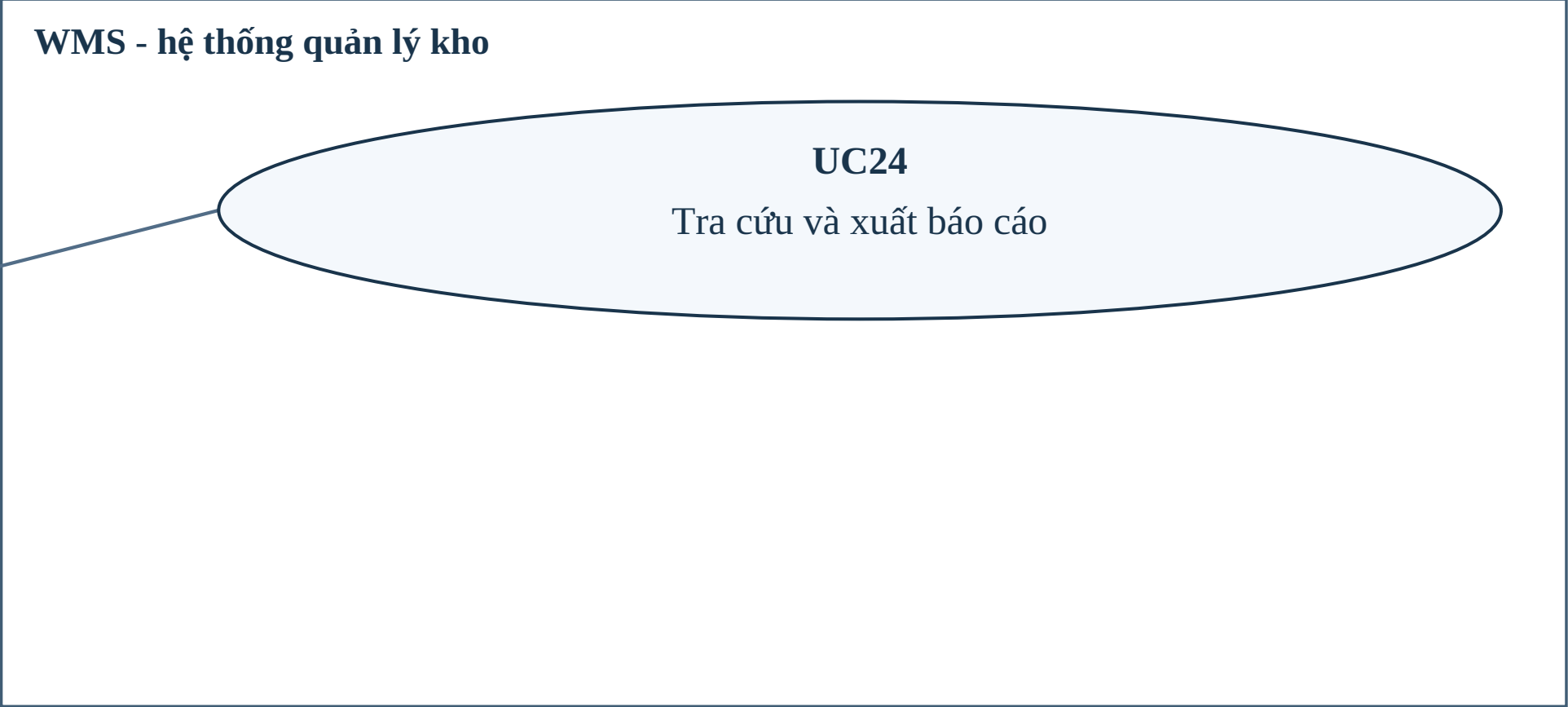
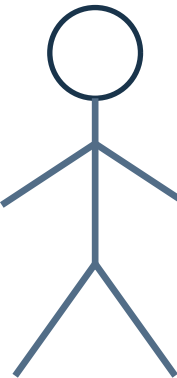
WMS / BA 2.0 / 27-09-2026 / DRAFT



Kiểm toán / Use case

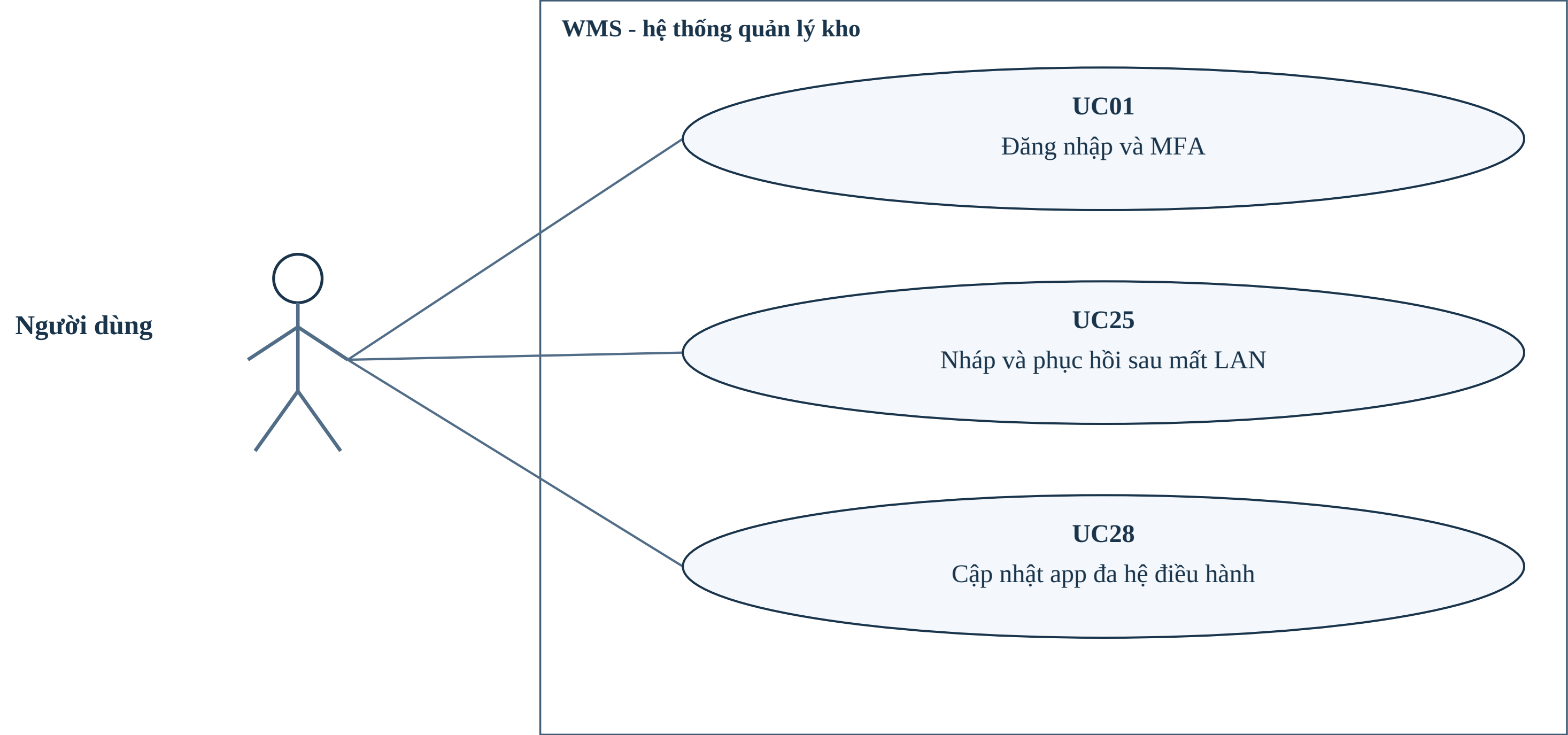
WMS / BA 2.0 / 27-09-2026 / DRAFT

Kiểm toán



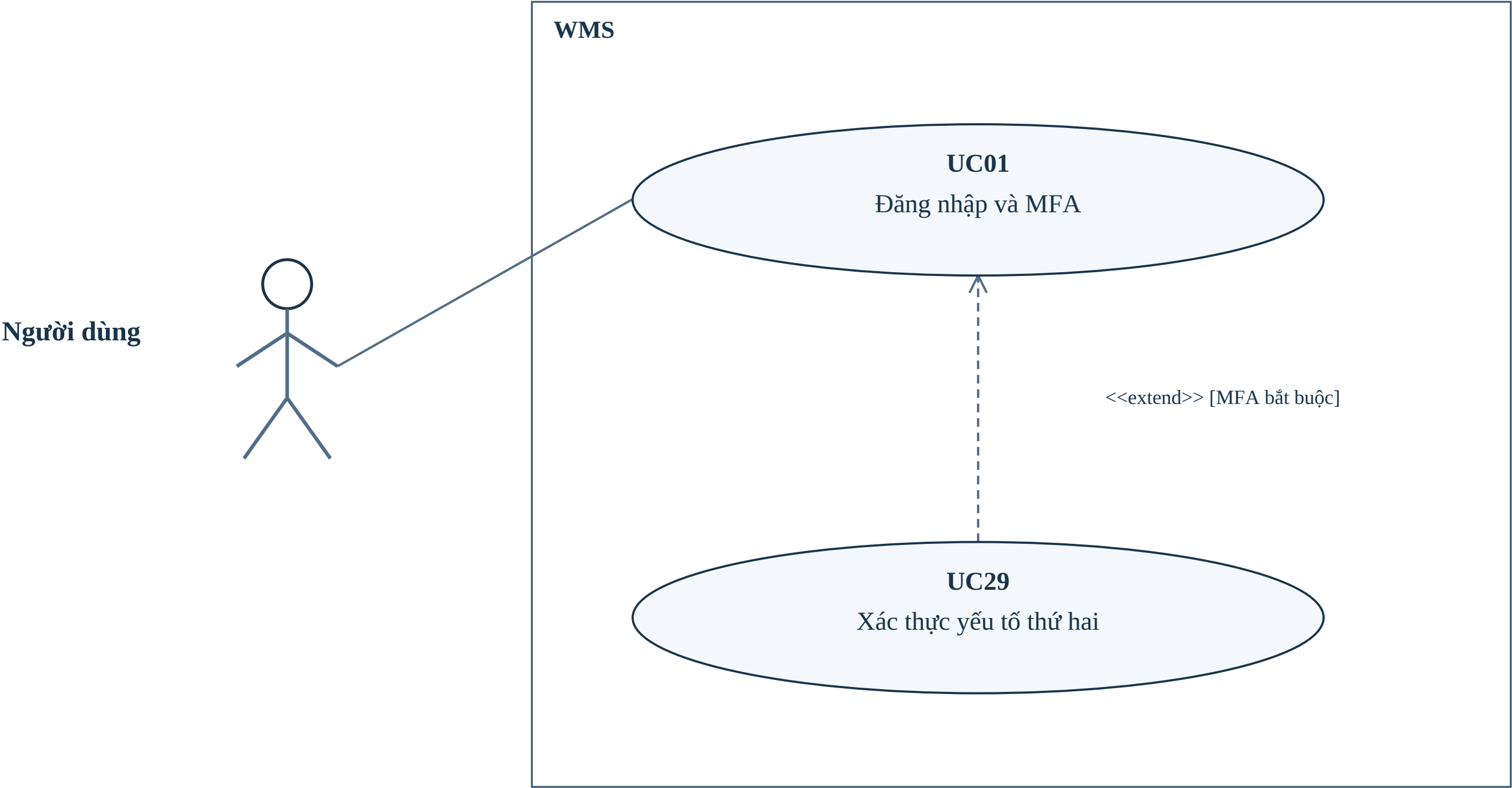
Người dùng / Tương tác chung

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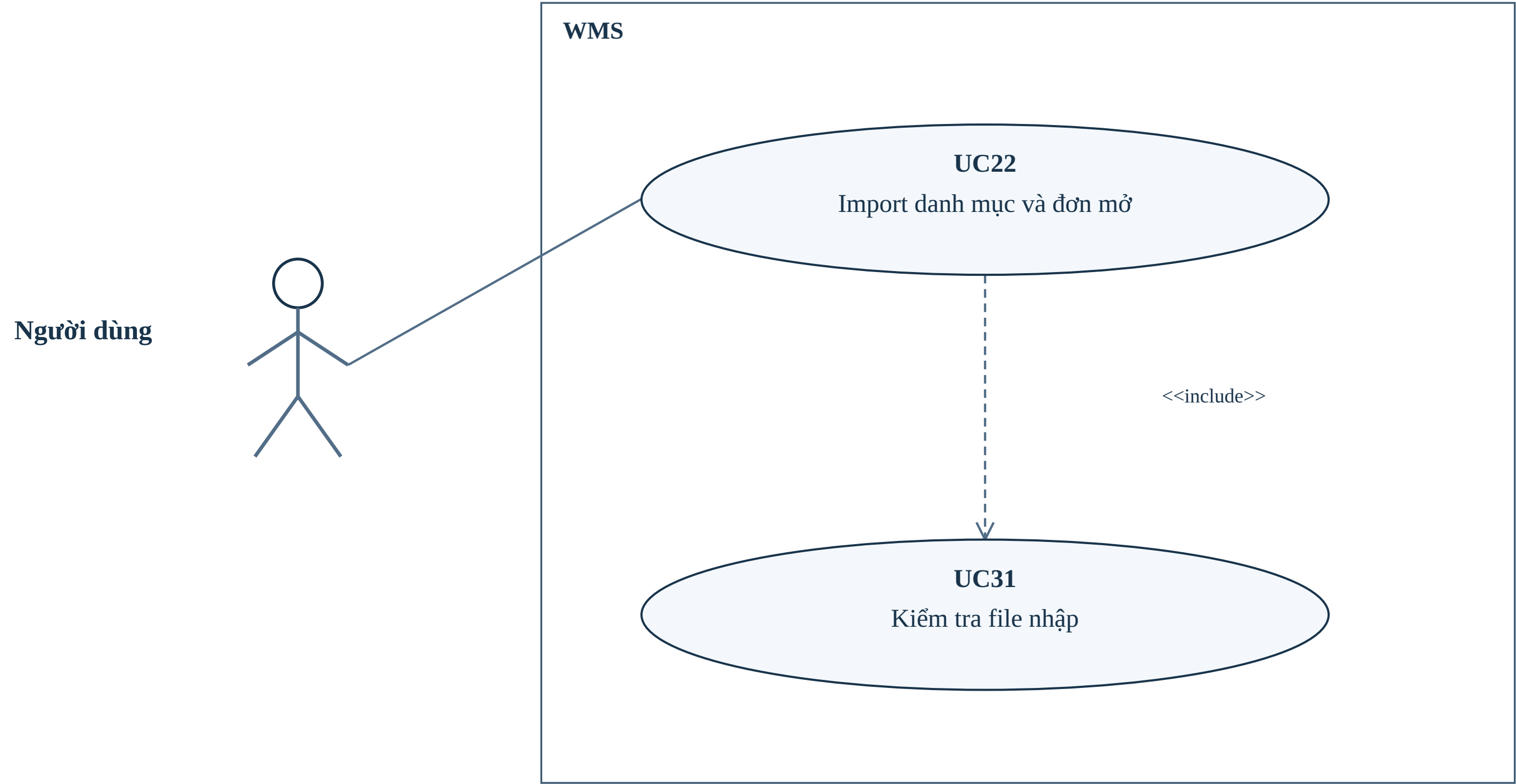
Quan hệ use case / extend

WMS / BA 2.0 / 27-09-2026 / DRAFT



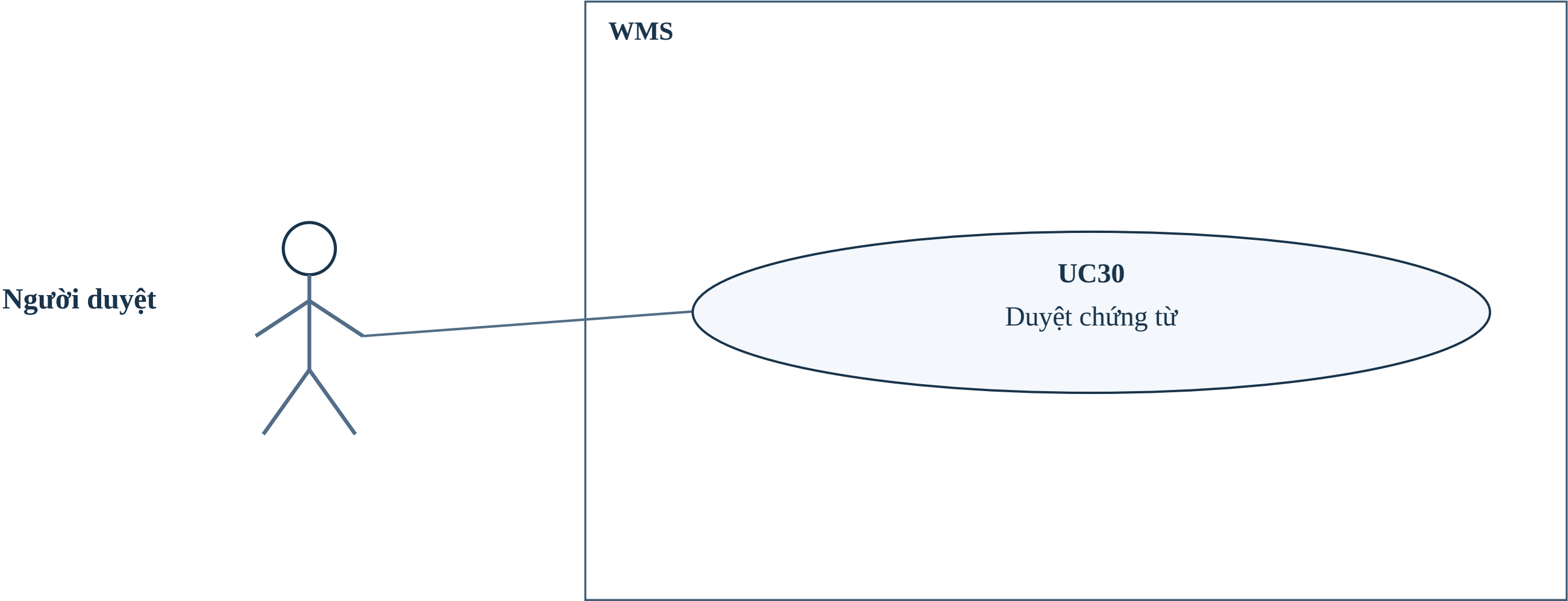
Quan hệ use case / include

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Duyệt chứng từ theo policy

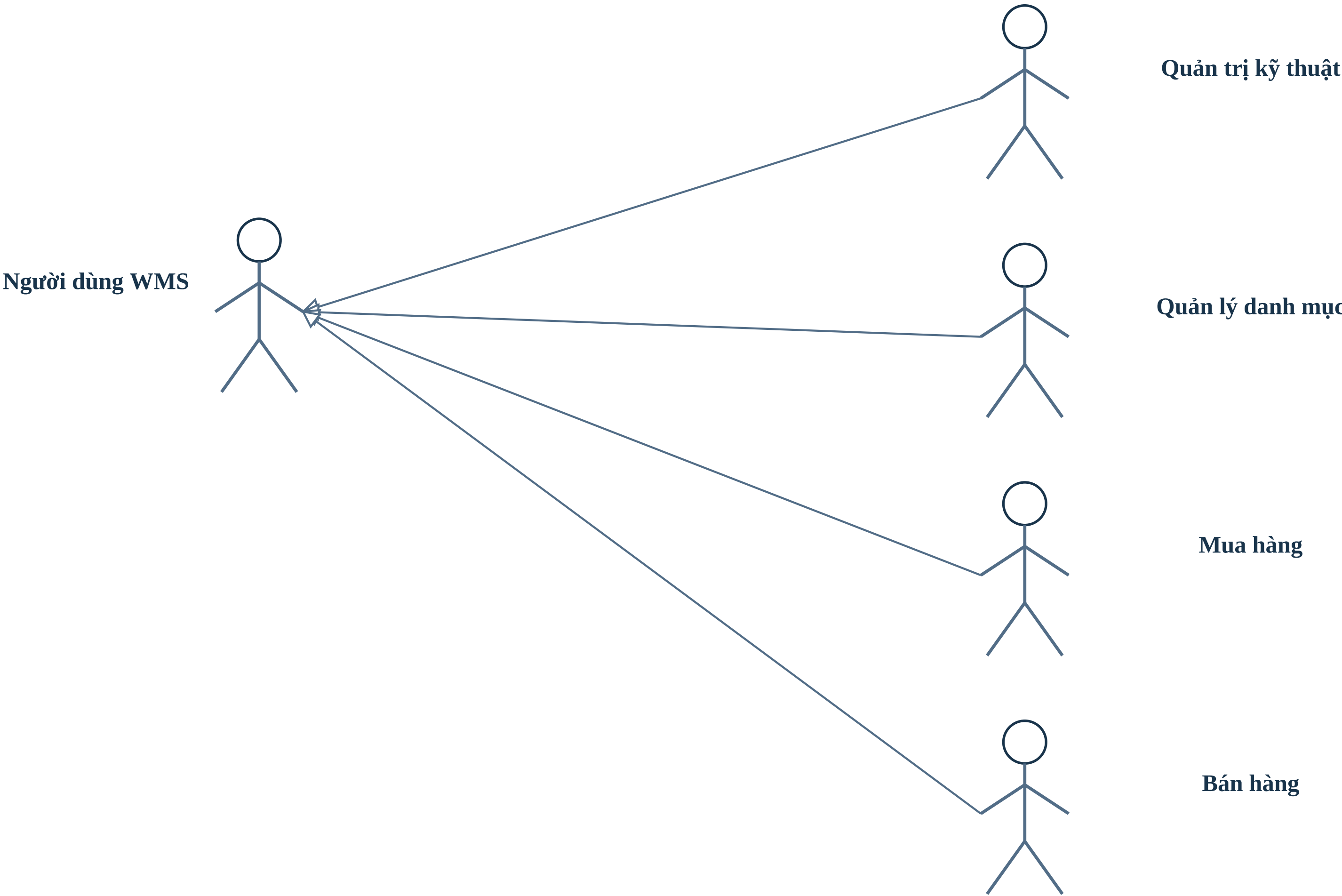
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UC30 là tác vụ độc lập; role và người được duyệt phụ thuộc loại phiếu, bước, kho và SOD.

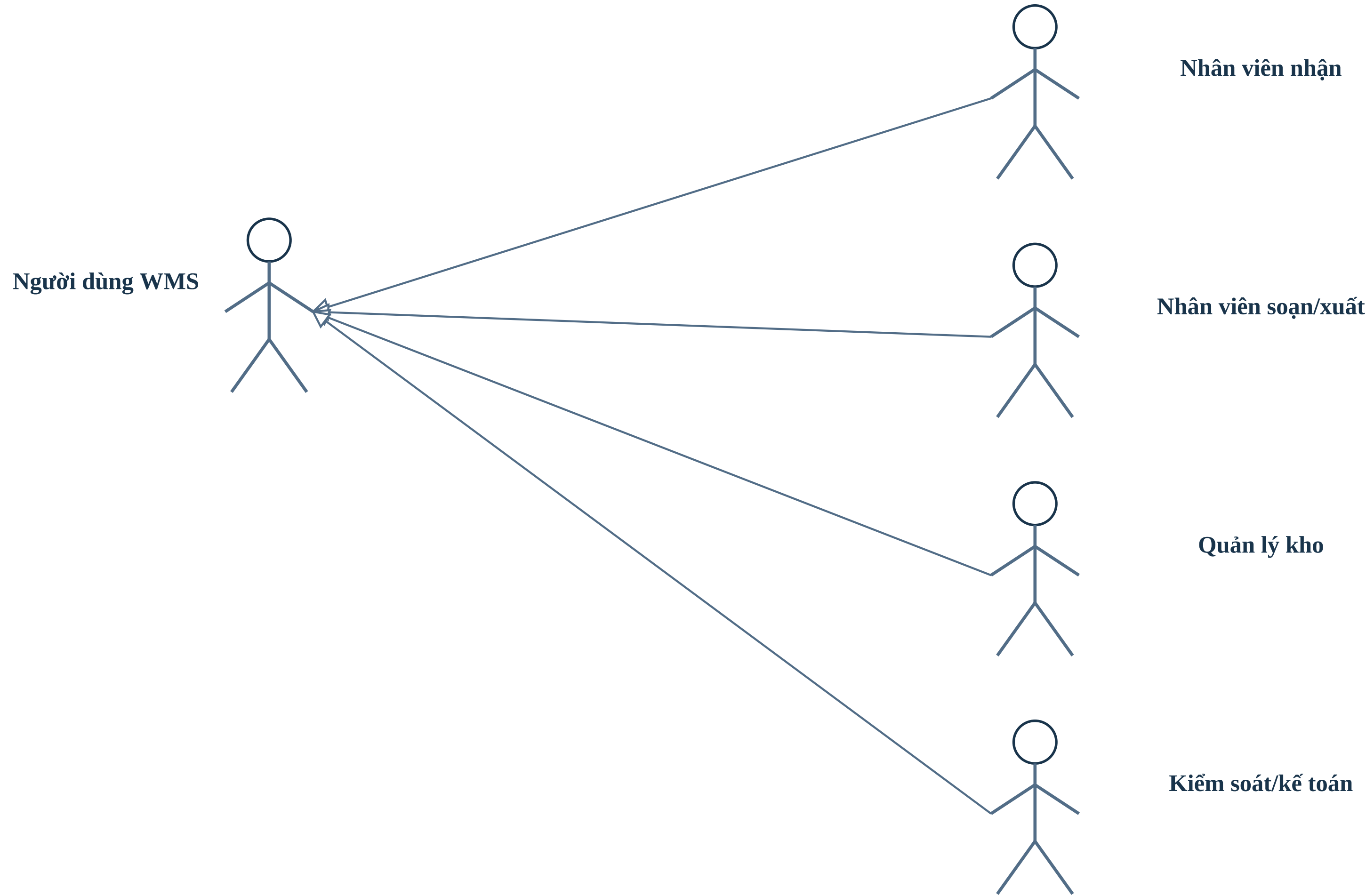
Tổng quát hóa vai trò người dùng

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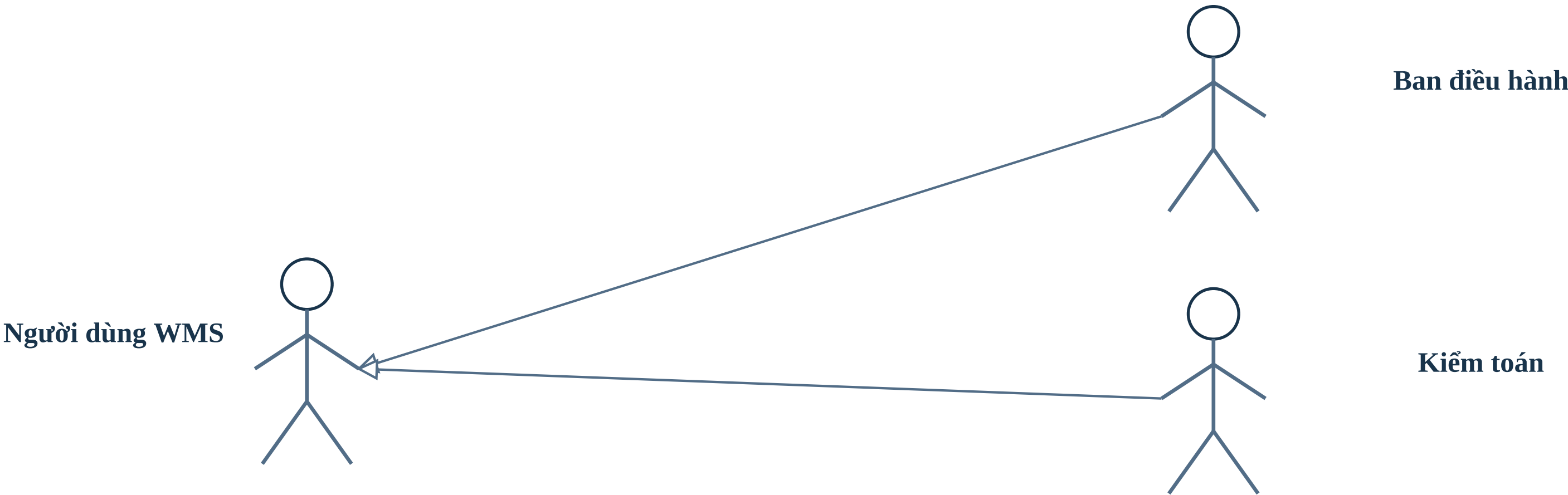
Tổng quát hóa vai trò người dùng

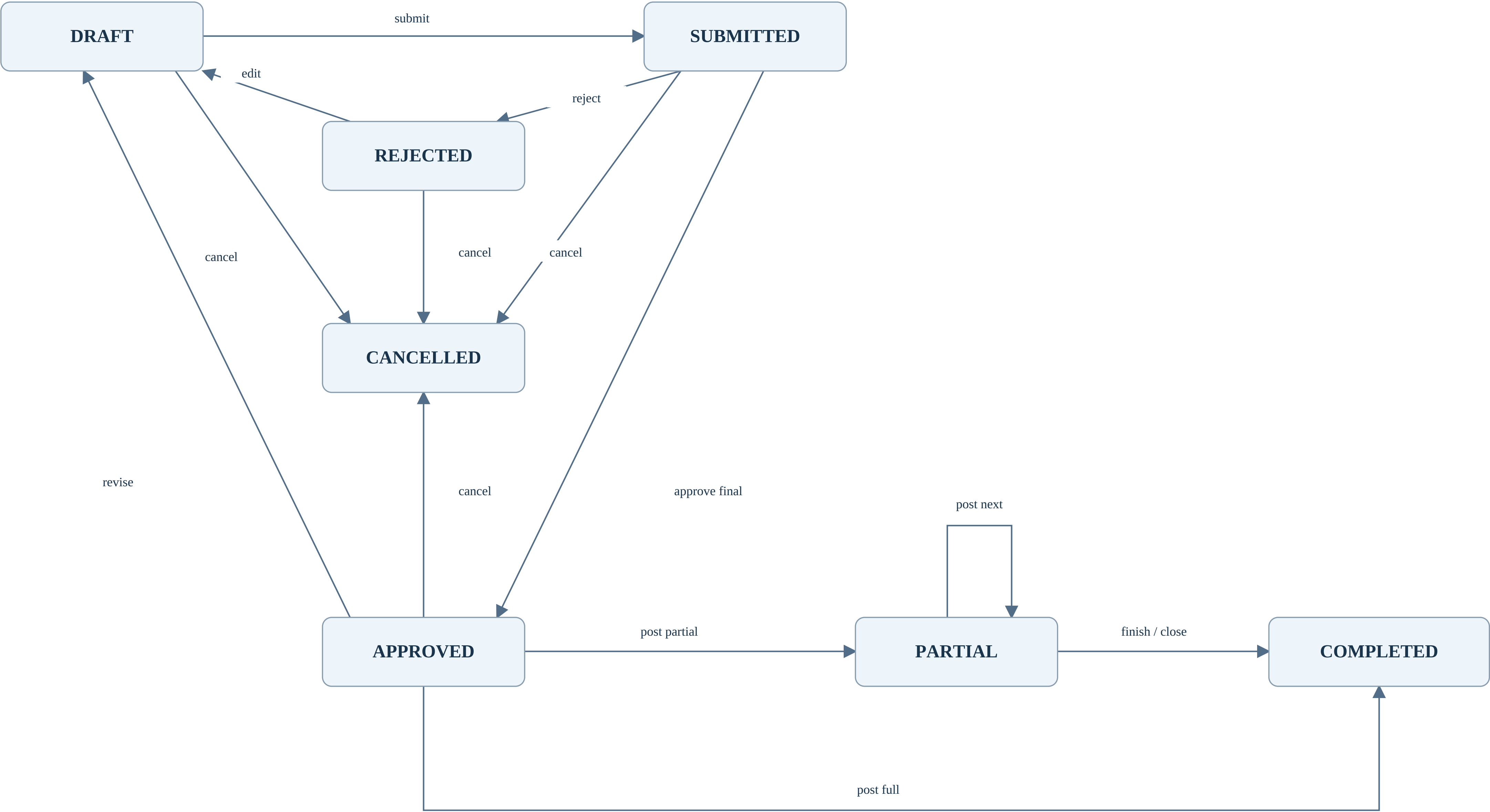
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Tổng quát hóa vai trò người dùng

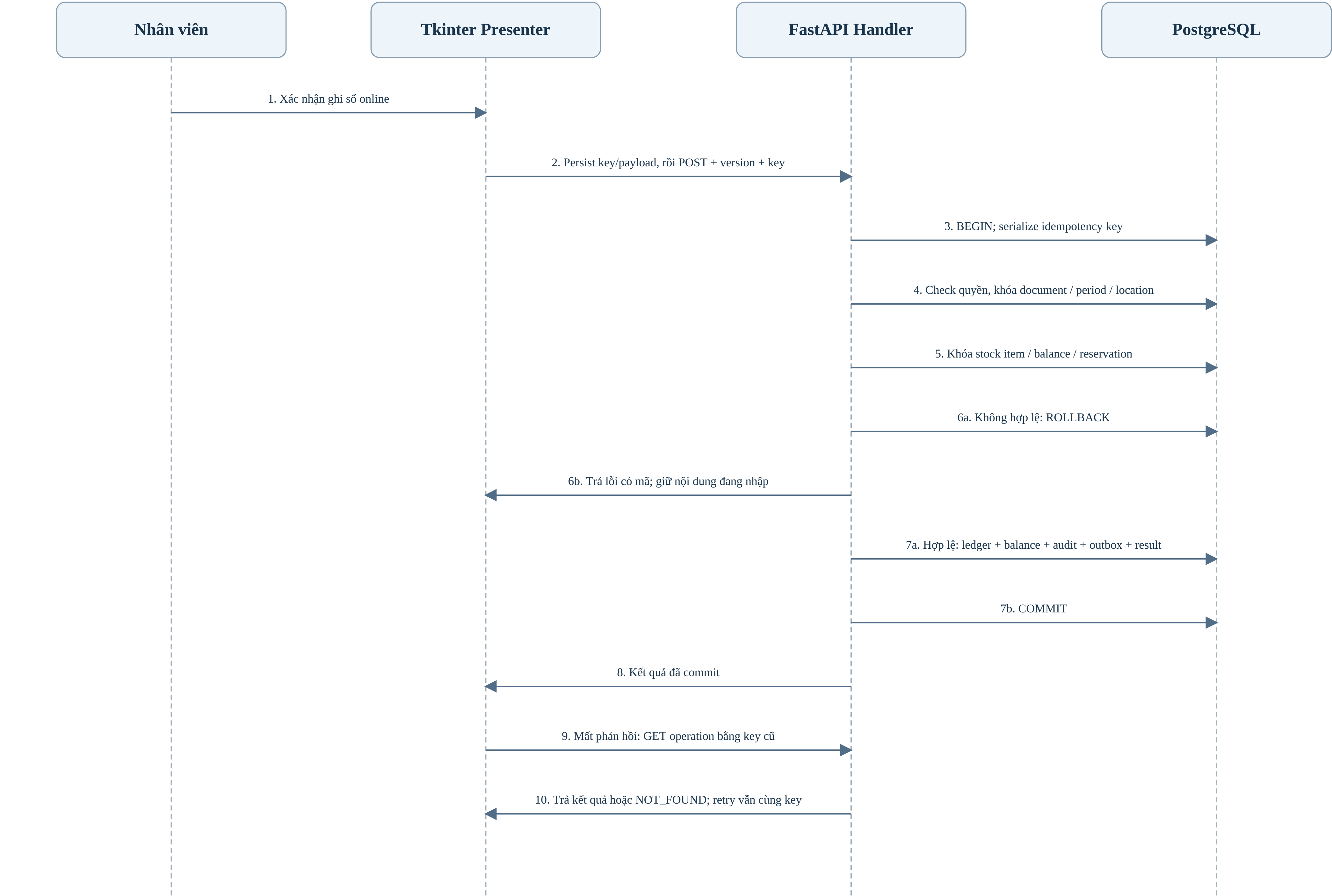
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Trình tự ghi sổ và xử lý timeout

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6a-6b là nhánh lỗi; 7a-8 là nhánh hợp lệ. Chỉ báo POSTED sau COMMIT. NOT_FOUND không cho phép đổi key.

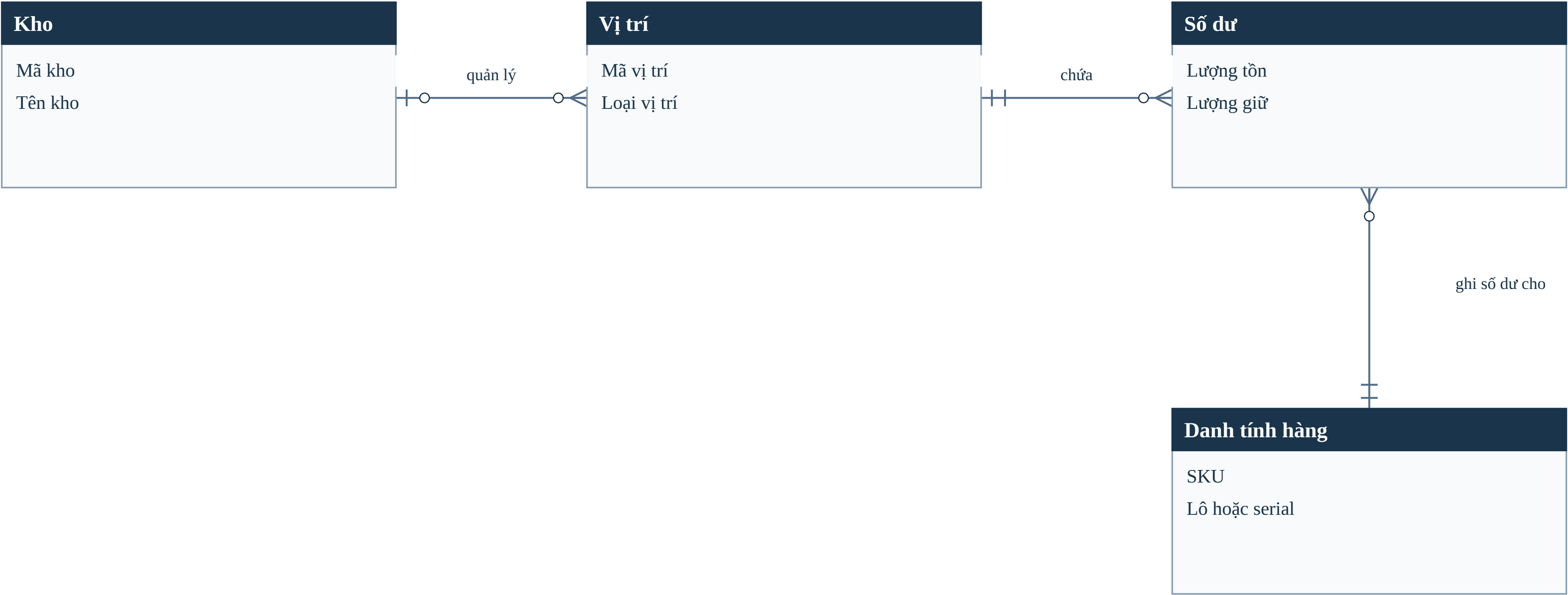
ERD khái niệm / Chứng từ và hàng hóa

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ERD khái niệm / Vị trí, hàng tồn và phát sinh

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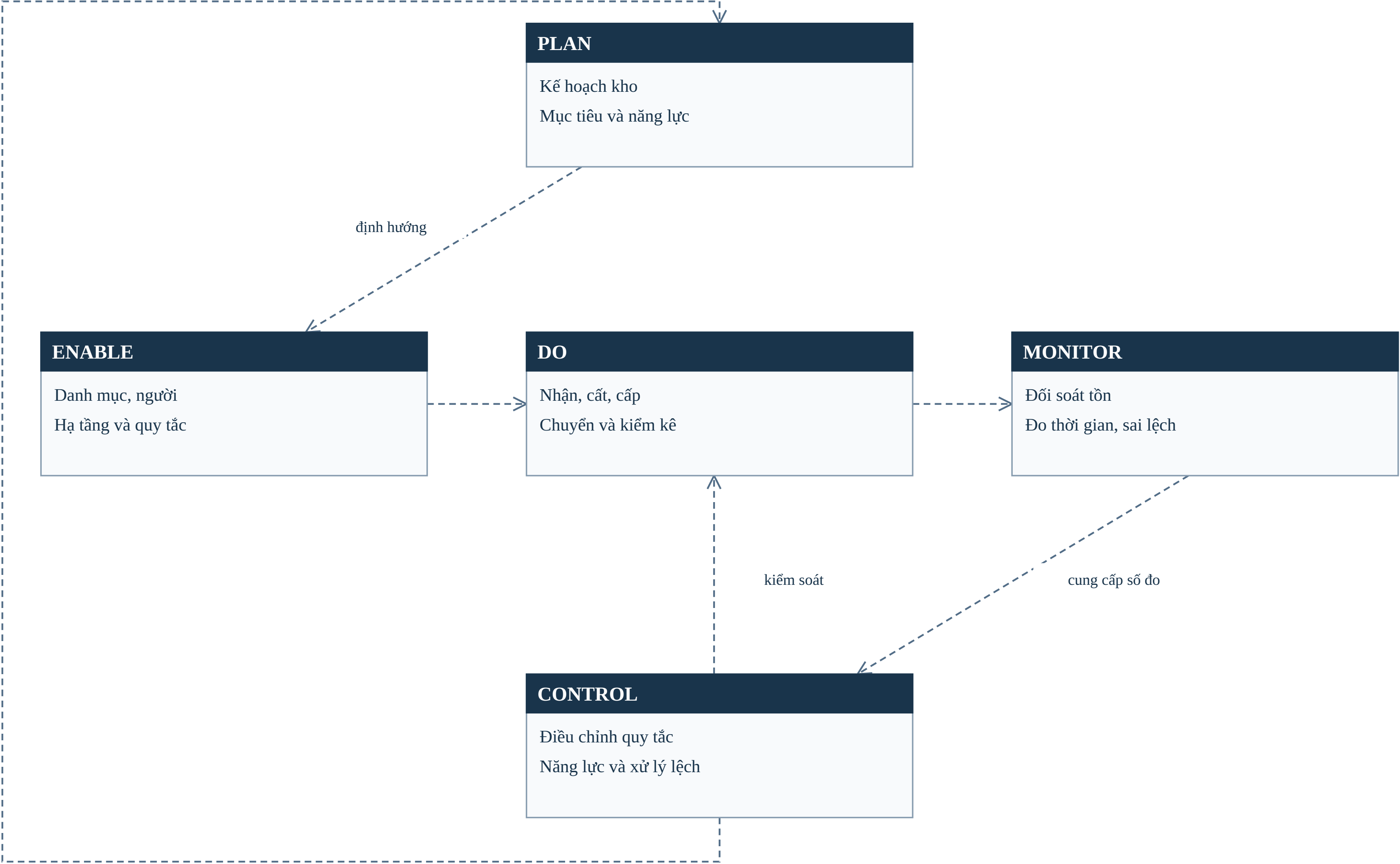
ERD khái niệm / Tài khoản, vai trò và quyền

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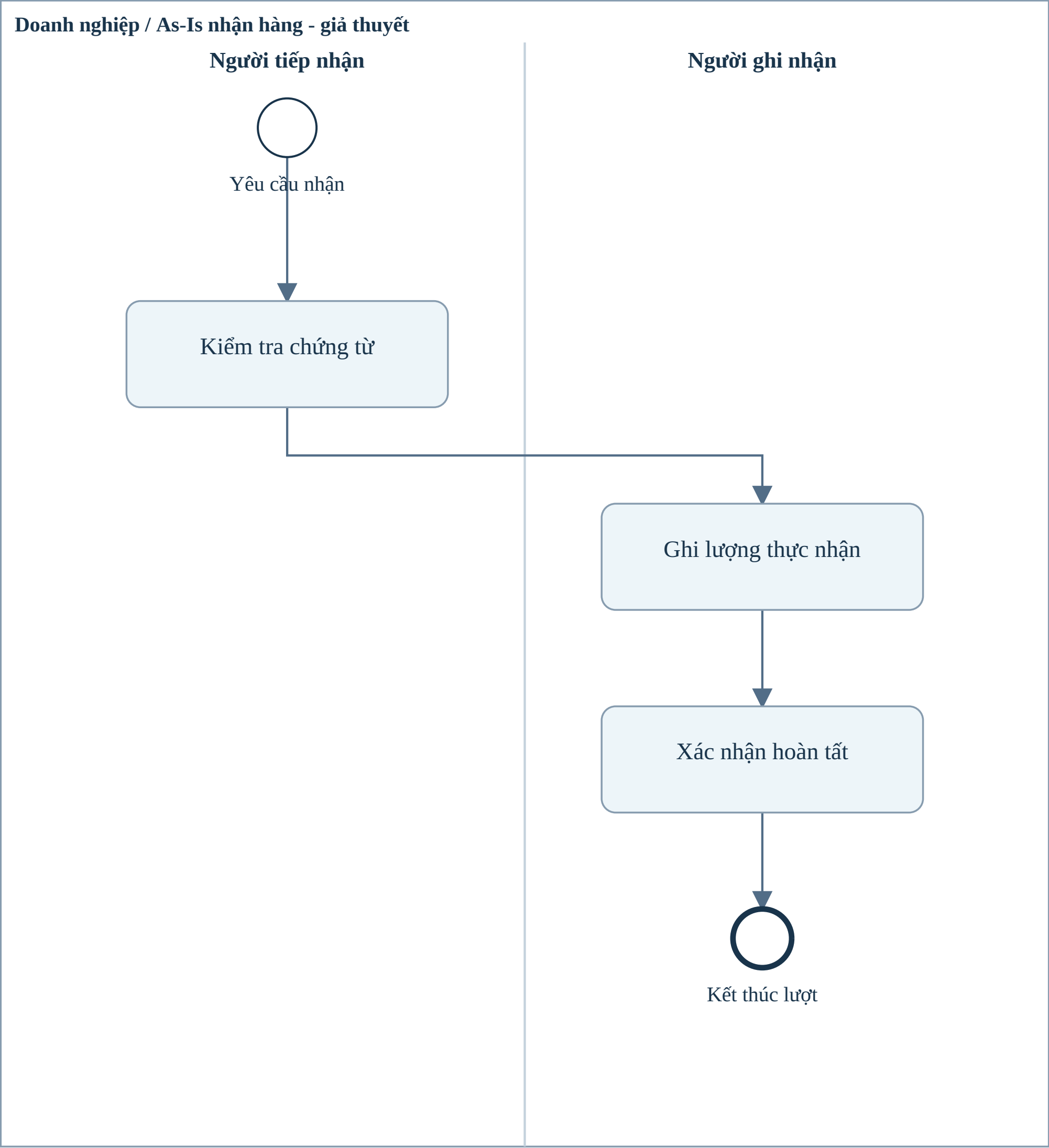
Business Activity Model / đề xuất

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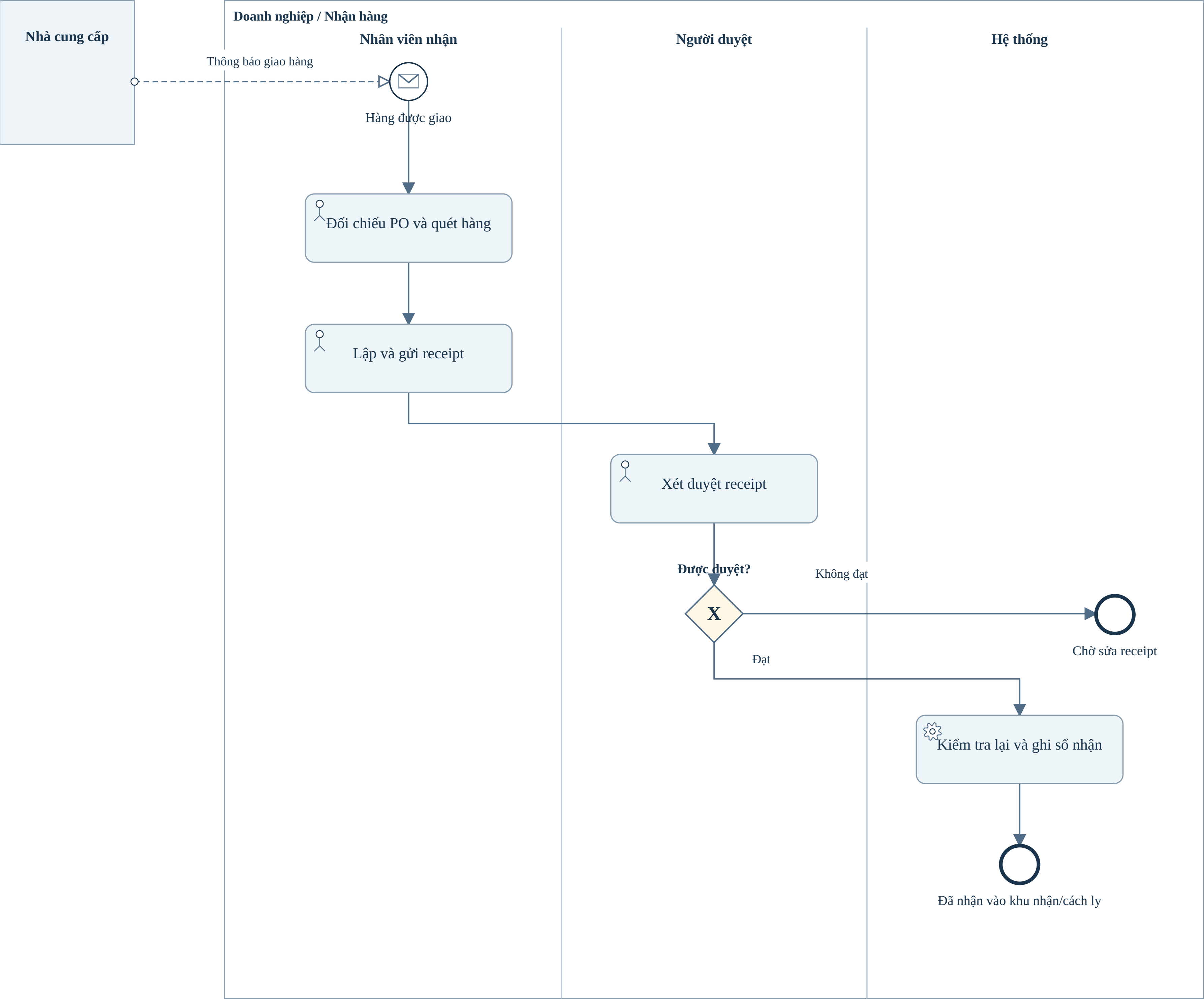
As-Is nhận hàng - giả thuyết

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Nhận hàng

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Xuất hàng

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